CHAPTER

3 Mole Concept

For NSEJS 2025

DALTON'S ATOMIC THEORY

Dalton's atomic theory was given by John Dalton in 1808. The main postulates of Dalton's atomic theory are:

1. Elements consist of indivisible, indestructible particles called atom.
2. Atoms of an element are identical to each other. They have the same mass and size.
3. Atoms of different elements differ in properties and have different masses and sizes.
4. Compounds are formed when atoms of different elements combine with each other in simple numerical ratio such as 1: 1, 2: 2, 2: 3 and so on.
5. Atoms cannot be created, destroyed or transformed into atoms of other elements.

The main failures of Dalton's atomic theory are:

1. It failed to explain how atoms of different elements differ from each other i.e., did not tell anything about structure of the atom.
2. It could not explain how and why atoms of different element combine with each other to form compound.
3. It failed to explain the nature of forces that bind together different atoms in a molecule.
4. It failed to explain Gay Lussac's law of combining volumes.
5. It did not make any distinction between ultimate particle of an element that takes part in reaction (atoms) and ultimate particle that has independent existence (molecules).

Prefixes Used With Units

The S.I. system recommends the multiples such as 10^3, 10^6, 10^9 etc. and fraction such as 10^{-3}, 10^{-6}, 10^{-9} etc., i.e. the powers are the multiples of 3. These are indicated by special prefixes. These along with some other fractions or multiples in common use, along with their prefixes are given below in Table and illustrated for length (m)

TABLE: SOME COMMONLY USED PREFIXES WITH THE UNITS

Prefix	Symbol	Multiplication Factor	Example
deci	d	10^{-1}	1 decimetre (dm) $= 10^{-1}$ m
centi	c	10^{-2}	1 centimetre (cm) $= 10^{-2}$ m
milli	m	10^{-3}	1 millimetre (mm) $= 10^{-3}$ m
micro	μ	10^{-6}	1 micrometre (μm) $= 10^{-6}$ m
nano	n	10^{-9}	1 nanometre (nm) $= 10^{-9}$ m
pico	p	10^{-12}	1 picometre (pm) $= 10^{-12}$ m
femto	f	10^{-15}	1 femtometre (fm) $= 10^{-15}$ m
atto	a	10^{-18}	1 attometre (am) $= 10^{-18}$ m
zepto	z	10^{-21}	1 zeptometre (zm) $= 10^{-21}$ m
yocto	y	10^{-24}	Yocto $= 10^{-24}$ m
deca	da	10	1 dekametre(dam) $= 10^{1}$ m
hecto	h	10^{2}	1 hectometre (hm) $= 10^{2}$ m
kilo	k	10^{3}	1 kilometre (km) $= 10^{3}$ m
mega	M	10^{6}	1 megamerte(Mm) $= 10^{6}$ m
giga	G	10^{9}	1 gigametre (Gm) $= 10^{9}$ m
tera	T	10^{12}	1 teramerte (Tm) $= 10^{12}$ m
peta	P	10^{15}	1 petametre (Pm) $= 10^{15}$ m
exa	E	10^{18}	1 exametre (Em) $= 10^{18}$ m
Zeter	Z	10^{21}	1 zetermetre (Zm) $= 10^{21}$
yotta	Y	10^{24}	Yotter $= 10^{24}$ m

Significant Figures:

"They are total number of digits in any number including the last digit whose value is uncertain, called as significant digits.'

Significant Figures = All certain digits + One doubtful digit (±1)

The uncertainty in the experimental or the calculated values is indicated by mentioning the number of significant figures. Significant figures are meaningful digits which are known with certainty. The uncertainty is indicated by writing the certain digits and the last uncertain digit.

Precision: Precision refers to the closeness of various measurements for the same quantity.

Accuracy: Accuracy is the agreement of a particular value to the true value of the result.

There are certain rules for determining the number of significant figures. These are stated below:

(1) All non-zero digits are significant. For example in 285 cm, there are three significant figures and in 0.25 mL, there are two significant figures.

(2) Zeros preceding to first non-zero digit are not significant. Such zero indicates the position of decimal point. Thus, 0.03 has one significant figure and 0.0052 has two significant figures.

(3) Zeros between two non-zero digits are significant. Thus, 2.005 has four significant figures.

(4) Zeros at the end or right of a number are significant provided they are on the right side of the decimal point. For example, 0.200 g has three significant figures. But, if otherwise, the terminal zeros are not significant if there is no decimal point. For example, 100 has only one significant figure, but 100 has three significant figures and 100.0 has four significant figures. Such numbers are better represented in scientific notation. We can express the number 100 as 1×10^2 for one significant figure, 1.0×10^2 for two significant figures and 1.00×10^2 for three significant figures.

(5) Counting numbers of objects, for example, 2 balls or 20 eggs, have infinite significant figures as these are exact numbers and can be represented by writing infinite number of zeros after placing a decimal i.e., 2 = 2.000000 or 20 = 20.000000 In numbers written in scientific notation, all digits are significant e.g., 4.01×10^2 has three significant figures, and 8.256×10^{-3} has four significant figures.

(a) Atomic mass

Relative Atomic Mass:

One of the most important concepts which come out from Dalton's atomic theory was that of relative atomic mass or relative atomic weight. This is done by expressing mass of one atom with respect to a fixed standard. Dalton used hydrogen as the standard (H = 1). Later on oxygen (O = 16) replaced hydrogen as the reference. Therefore, relative atomic mass is given as

$$\text{Relative atomic mass (R.A.M)} = \frac{\text{Mass of one atom of an element}}{\text{mass of one hydrogen atom}}$$

$$= \frac{\text{Mass of one atom of an element}}{\frac{1}{16}\times\text{mass of one oxygen atom}}$$

- The present standard unit which was adopted internationally in 1961, is based on the mass of one carbon-12 atom.

Atomic Mass Unit (or amu):

The atomic mass unit (amu) is equal to one twelfth $\left(\frac{1}{12}\right)$ the mass of one atom of carbon-12 isotope.

$$\therefore \text{ 1 amu} = \frac{1}{12}\times\text{mass of one C}-12\text{ atom}$$

$\simeq$ mass of one nucleon in C-12 atom. = 1.66×10^{-24} gm or 1.66×10^{-27} kg

- one amu is also called one Dalton (Da).

Today, amu has been replaced by 'U' which is known as unified mass

Now the relative atomic mass is given as

$$\text{Relative atomic mass} = \frac{\text{mass of one atom of the element}}{\frac{1}{12} \times \text{mass of one C} - 12 \text{ atom}}$$

$$\text{R.A.M.} = \frac{\text{Atomic mass}}{1 \text{ amu}}$$

Atomic mass = R.A.M × 1 amu

$$\text{Relative molecular mass} = \frac{\text{mass of one molecule of the subs tance}}{\frac{1}{12} \times \text{mass of one C} - 12 \text{ atom}}$$

$\therefore$ Molecular mass = Relative molecular mass × 1 amu

Atomic mass: It is the mass of 1 atom of a substance it is expressed in amu.

ATOMIC MASS = R.A.M × 1 amu

Mind it: Relative atomic mass is nothing but the number of nucleons present in the atom.

Solved Examples

Example : Find the relative atomic mass of 'O' atom and its atomic mass.

Solution : The number of nucleons present in 'O' atom is 16.

$\therefore$ relative atomic mass of 'O' atom = 16.

Atomic mass = R.A.M × 1 amu = 16 × 1 amu = 16 amu

Average / Mean Atomic Mass:

The weighted average of the isotopic masses of the element's naturally occurring isotopes.

Mathematically, average atomic mass of X (A_x) = $\frac{a_1x_1 + a_2x_2 + ... + a_nx_n}{100}$

a_1 = atomic mass; x_1 % occurrence in nature

Example : Naturally occuring chlorine is 75% Cl^{35} which has an atomic mass of 35 amu and 25% Cl^{37} which has a mass of 37 amu. Calculate the average atomic mass of chlorine –

(A) 35.5 amu (B) 36.5 amu (C) 71 amu (D) 72 amu

Solution : (1) Average atomic mass = $\frac{75 \times 35 + 25 \times 37}{100}$ = 35.5 amu

Mind it: (a) In all calculations we use this mass.

(b) In periodic table we report this mass only.

Gram Atomic Mass:

The atomic mass of an element expressed in gram is called gram atomic mass of the element.

or

It is also defined as mass of 6.02×10^{23} atoms.

or

It is also defined as the mass of one mole of atoms.

For example for oxygen atom:

Atomic mass of 'O' atom = mass of one 'O' atom = 16 amu

gram atomic mass = mass of 6.02×10^{23} 'O' atoms

$= 16 \text{ amu} \times 6.02 \times 10^{23}$

$= 16 \times 1.66 \times 10^{-24} \text{ g} \times 6.02 \times 10^{23} = 16 \text{ g}$

($\because 1.66 \times 10^{-24} \times 6.02 \times 10^{23} \simeq 1$)

Solved Examples

Example : How many atoms of oxygen are there in 16 g oxygen.

Solution : Let x atoms of oxygen are present

So, $16 \times 1.66 \times 10^{-24} \times x = 16 \text{ g}$

$$x = \frac{1}{1.66 \times 10^{-24}} = N_A$$

(b) Molecular Mass

MOLECULE: It is the smallest particle of matter which is capable of independent existence.

- A molecule is generally an assembly of two or more tightly bonded atoms.
- Molecules are of two type on the basis of elemental atoms.

(i) Homo atomic molecules: Molecules of an element containing one type of atoms only.
Example: O_2, H_2, Cl_2 etc.

(ii) Heteroatomic molecules: Molecules of compounds containing more than one type of atoms.
Example: NH_3, H_2O, CH_4 etc.

Solved Examples

Example : The molecular mass of H_2SO_4 is 98 amu. Calculate the number of moles of each element in 294 g of H_2SO_4.

Solution : Gram molecular mass of $H_2SO_4 = 98$ gm

$$\text{moles of } H_2SO_4 = \frac{294}{98} = 3 \text{ moles}$$

	H_2SO_4	H	S	O
	One molecule	2 atom	One atom	4 Atom
	$1 \times N_A$	$2 \times N_A$ atoms	$1 \times N_A$ atoms	$4 \times N_A$ atoms
$\therefore$	One mole	2 mole	One mole	4 mole
$\therefore$	3 mole	6 mole	3 mole	12 mole

Gram Molecular Mass:

The molecular mass of a substance expressed in gram is called the gram-molecular mass of the substance.

or

It is also defined as mass of 6.02×10^{23} molecules

or

It is also defined as the mass of one mole of molecules.

For example for 'O_2' molecule:
Molecular mass of 'O_2' molecule = mass of one 'O_2' molecule
= 2 × mass of one 'O' atom = 2 × 16 amu = 32 amu
gram molecular mass = mass of 6.02×10^{23} 'O_2' molecules = 32 amu × 6.02×10^{23}
= $32 \times 1.66 \times 10^{-24}$ gm × 6.02×10^{23} = 32 gm

Mean Molar Mass or Molecular Mass:

The average molar mass of the different substance present in the container = $\frac{n_1M_1 + n_2M_0 + ...n_nM_n}{n_1 + n_2 + ...n_n}$

n_1 = Number of moles; M = Molar mass

Solved Examples

Example : The molar composition of polluted air is as follows:

Gas	At. wt.	mole percentage composition
Oxygen	16	16%
Nitrogen	14	80%
Carbon dioxide	–	03%
Sulphurdioxide	–	01%

What is the average molecular weight of the given polluted air? (Given, atomic weights of C and S are 12 and 32 respectively.

Solution : $M_{avg} = \frac{\sum_{j=1}^{j=n} n_j M_j}{\sum_{j=1}^{j=n} n_j}$ Here $\sum_{j=1}^{j=n} n_j = 100$

$\therefore \quad M_{avg} = \frac{16 \times 32 + 80 \times 28 + 44 \times 3 + 64 \times 1}{100}$

$= \frac{512 + 2240 + 132 + 64}{100} = \frac{2948}{100} = 29.48$ Ans.

Density:

It is of two type.

- Absolute density
- Relative density

For Liquid and Solids

- Absolute density = $\frac{\text{mass}}{\text{volume}}$
- Relative density or specific gravity = $\frac{\text{density of the substance}}{\text{density of water at } 4^oC}$

  We know that density of water at 4ºC = 1 g/ml.

For Gases:

- Absolute density (mass/volume) = $\frac{\text{Molar mass}}{\text{Molar volume}}$

  where P is pressure of gas, M = mol. wt. of gas, R is the gas constant, T is the temperature.

RELATIVE DENSITY OR VAPOUR DENSITY

Vapour density is defined as the density of the gas with respect to hydrogen gas at the same temperature and pressure.

Vapour density = $\frac{d_{gas}}{d_{H_2}} = \frac{PM_{gas}/RT}{PM_{H_2}/RT}$

V.D. = $\frac{M_{gas}}{M_{H_2}} = \frac{M_{gas}}{2}$

M_{gas} = 2 V.D.

Relative density can be calculated w.r.t. to other gases also.

Solved Examples

Example : What is the V.D. of SO_2 with respect to CH_4

Solution : V.D. = $\frac{M.W.SO_2}{M.W.CH_4}$ V.D = $\frac{64}{16}$ = 4

Example : 7.5 litre of the particular gas at S.T.P. weighs 16 gram. What is the V.D. of gas

Solution : 7.5 litre = 16 gram

moles = $\frac{7.5}{22.4} = \frac{16}{M}$ M = 48 gram V.D. $\frac{48}{2}$ = 24

LAWS OF CHEMICAL COMBINATION

(A) Law of conservation of mass/Law of indestructibility of matter

Given by – Lavoisier

Tested by – Landolt

Statement: According to law of conservation of mass in all physical & chemical changes total mass of the system remains constant. or

In a physical or chemical change mass neither be created nor destroyed.

i.e. Total mass of reactant = Total mass of the product

→ This relationship hold good when reactant are completely converted into products.

→ If reactant are not completely consume then the relationship will be:

Total mass of reactant = Total mass of product + Mass of unreacted reaction

Exception: Nuclear reaction are exception of law of conservation of mass. In nuclear reaction mass + energy is conserved.

Modern View: According to the modern views, the law of conservation of mass is not always valid. The law hold good only in case of such chemical reactions where there is no evolution of heat or light. According to Einstein's theory of relativity, mass and energy are interconvertible according to the equation.

$E = mc^2$

where E = Energy content of the body

m = Mass of the body.

C = Velocity of light (3×10^{10} cm/sec)

E = (Change in mass) × c^2

As c is very large, change in mass is found to be extremely small in normal reactions.

During chemical processes, the loss of mass is negligible. But in nuclear reactions, tremendous amount of energy is evolved. Consequently, the change in mass is quite significant. Thus, it is clear that the law of conservation of mass and law of conservation of energy are two ways of looking at the same law.

Therefore, combining the two we get general law known as law of conservation of mass energy. It states that, Mass and energy are inter convertible. But the total sum of mass and energy of the system remains constant.

Solved Examples

Example : What weight of $BaCl_2$ would react with 24.4 g of sodium sulphate to produce 46.6 g of barium sulphate and 23.4 g of sodium chloride ?

Solution : Barium chloride and sodium sulphate react to produce barium sulphate and sodium chloride according to the equation:

$BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$

x g 24.4 g 46.6 g 23.4 g

Let the weight of $BaCl_2$ be x g. According to law of conservation of mass:

Mass of reactants = Mass of products

Mass of reactants = (x + 24.4) g

Mass of products = (46.6 + 23.4) g

Equating the two masses

$x + 24.4 = 46.6 + 23.4$

$x = 46.6 + 23.4 - 24.4$ or $x = 45.6$ g

hence, the weight of $BaCl_2$ is 45.6 g

Example : 10 g of $CaCO_3$ on heating gives 4.4 g of CO_2 then determine weight of produced CaO in quintal.

Solution : Total mass of reactant = 10 g

Mass of CO_2 = 4.4 g

Mass of produced CaO = x

According to law of conservation of mass

$10 = 4.4 + x$

$10 - 4.4 = x$

$x = 5.6$ g

$\therefore$ 1 quintal = 100 kg

$\therefore$ 1 Kg = 1000 g

$= 5.6\text{g} \times \frac{\text{kg}}{1000} = 5.6 \times 10^{-3} \times \text{Kg}$

$= 5.6 \times 10^{-3} \times \frac{1}{100}$ quintal $= 5.6 \times 10^{-5}$ quintal

Law of definite proportions/constant composition:

Given by $\rightarrow$ Joseph Proust

Statement: A chemical compounds always contains same elements combined together in same proportion by mass. i,e, chemical compound has a fixed composition & it does not depends on the method of its preparation or the source from which it has been obtained.

For example, molecule of ammonia always has the formula NH_3. That is one molecule of ammonia always contains. One atom of nitrogen and three atoms of hydrogen or 17.0 g of NH_3 always contains 14.0 g of nitrogen and 3 g of hydrogen.

For Example: CO_2 can be obtained by using any one of the following method.

(1) By burning of carbon in the presence of O_2:

$C + O_2 \rightarrow CO_2$

(2) By decomposition of $CaCO_3$

$CaCO_3 \rightarrow CaO + CO_2 \uparrow$

(3) $CaCO_3 + 2\,HCl \rightarrow CaCl_2 + H_2O + CO_2 \uparrow$

In CO_2 wt. ratio = C: O

= 12: 32 = 3: 8

Whatever sample of CO_2 is taken it is observed that carbon & oxygen are always combined in the ratio of 12: 32 i.e. 3: 8.

Solved Examples

Example : Weight of copper oxide obtained by treating 2.16 g of metallic copper with nitric acid and subsequent ignition was 2.70 g. In another experiment, 1.15 g of copper oxide on reduction yielded 0.92 g of copper. Show that the results illustrate the law of constant composition.

Solution : Ist experiment

Mass of copper oxide = 2.70 g

$\therefore$ Mass of oxygen = 2.70 – 2.16 = 0.54 g

The ratio of the masses of copper and oxygen in the Ist experiment is 2.16: 0.54 = 4 : 1

IInd experiment

Mass of copper oxide = 1.15 g; Mass of copper = 0.92 g

$\therefore$ Mass of oxygen = 1.15 – 0.92 = 0.23 g

The ratio of the masses of copper and oxygen in the IInd experiment is 0.92: 0.23 = 4 : 1

Thus, the ratio of the masses of copper and oxygen in the two experiment is the same, i.e. 4 : 1.

Hence, the law of constant composition is proved.

Example : Illustrate the law of definite proportions from the following data:

(a) 0.32 g of sulphur on burning in air produced 224 ml of SO_2 at NTP.

(b) Sulphur dioxide obtained by the decomposition of sulphate contains 50% of sulphur.

Solution : Molecular weight of SO_2 = 32 + 32 = 64

224 ml of SO_2 at NTP will weight = $\frac{224 \times 64}{22400} = 0.64$

(a) In first Experiment

64 g of SO_2 contain = 32 g sulphur

$\therefore$ 0.64 g of SO_2 will contain = $\frac{0.64 \times 32}{64} = 0.32$ g sulphur

Similarly, 64 g of SO_2 contain = 32 g oxygen

0.64 g of SO_2 will contain = $\frac{0.64 \times 32}{64} = 0.32$ Oxygen

(b) As sulphur and oxygen are equal in amount in SO_2

Percentage of sulphur in SO_2 = 50%; Percentage of oxygen in SO_2 = 50%

Thus, the percentage of sulphur and oxygen in both the experiments is same. This illustrates the law of constant composition.

Law of multiple proportion: Given by → Daltons

Statement: When two elements combine to form more than one compound than the different mass of one element which combine with a fixed mass of other element bear a simple ratio to one another.

The following examples will illustrate.

(i) Nitrogen and oxygen combine to form five oxides, which are:

Nitrous oxide (N_2O), nitric oxide (NO), nitrogen trioxide(N_2O_3), nitrogen tetraoxide (N_2O_4) and nitrogen pentoxide (N_2O_5).

weights of oxygen which combine with the fixed weight of nitrogen in these oxides are calculated as under:

Oxide Ratio of weights of nitrogen and oxygen in each compound

N_2O	28: 16 = 14: 8
NO	14: 16 = 14: 16
N_2O_3	28: 48 = 14: 24
N_2O_4	28: 64 = 14: 32
N_2O_5	28: 80 = 14: 40

Number of parts by weight of oxygen which combine with 14 parts by weight of nitrogen from the above are 8,16,24,32 and 40 respectively. Their ratio is 1: 2: 3: 4: 5, which is a simple ratio. Hence the law illustrated.

(ii) Sulphur combines with oxygen to from two oxides, viz SO_2 and SO_3. The weights of oxygen which combine with a fixed weight, i.e. 32 parts by weight of sulphur in two oxides are in the ratio of 32:48 or 2: 3 which is simple ratio. Hence the law of multiple proportions is illustrated.

Solved Examples

Example : On analysis it was found that the black oxide of copper and the red oxide of copper contain 79.9% and 88.8% metal respectively. Establish the law of multiple proportions with the help of this data.

Solution : In the black oxide,

79.9 g copper combines with (100 – 79.9), i.e. 20.1 g oxygen

$\therefore$ In red oxide 88.8 g copper will combine with 100 – 88.8 = 11.2 g

$\therefore$ According to red oxide 79.9 copper will combine with $\frac{11.2 \times 79.9}{88.8}$ = 10.08 g oxygen

Thus the weights of oxygen that combine with the same 79.9 g copper are 20.1 g and 10.08 respectively.

These are in the ratio 20.1: 10.08 = 2: 1

It is a simple whole number ratio. Hence, the law of multiple proportions is established.

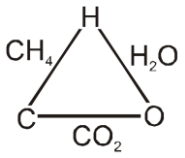
Law of reciprocal proportion:

Given by → Richter.

The ratio of the weights of two elements A and B which combine separately with fixed weight of the third element C is either the same or some simple multiple of this ratio of the weights in which A and B combine directly with each other.

This law is illustrated with the help of the following examples:

The elements C and O combine separately with the third element H to form CH_4 and H_2O and they combine directly with each other to form CO_2 as shown in the adjoining figure.



In CH_4, 12 parts by weight of carbon combine with 4 parts by weight of hydrogen. In H_2O, 2 parts by weight of hydrogen combine with 16 parts by weights of oxygen. Thus the weights of C and O which combine with fixed weight of hydrogen (say 4 parts of weight) are 12 and 32, i.e. they are in the ratio 12: 32 or 3: 8.

Now in CO_2, 12 parts by weight of carbon combine directly with 32 parts by weight of oxygen, i.e. they combine directly in the ratio 12: 32 or 3: 8 which is the same as the first ratio.

Solved Examples

Example : Copper sulphide contains 66.6% Cu, copper oxide contains 79.9% copper and sulphur trioxide contains 40% sulphur. Show that these data illustrate law of reciprocal proportions.

Solution : In copper sulphide,

Cu: S mass ratio is 66.6: 33.4

In sulphur trioxide,

O: S mass ratio is 60: 40

Now in copper sulphide

33.4 parts of sulphur combines with Cu = 66.6 parts

40.0 parts of sulphur combines with Cu.

$= \frac{66.6 \times 40}{33.4} = 79.8$ parts

Now ratio of the masses of Cu and O which combines with same mass (40 parts) of sulphur separately is 79.8: 60

Cu: O ratio by mass in CuO is 79.9: 20.1

Ratio I: Ratio II $= \frac{79.8}{60} \times \frac{20.1}{79.9} = 3: 1$

Which is simple whole number ratio.

Hence, law of reciprocal proportion is proved.

Gay–Lussac's law of gaseous Volume:

Given by $\rightarrow$ Gay Lussac

Statement: When gases react with each other in simple ratio of their volume & if product is also in gaseous state then the volume of the product is also bear a simple ratio to the volume of gaseous reactant when all volumes are measured under similar condition of temperature & pressure.

The following examples illustrate the law:

(i) One volume of hydrogen combines with one volume of chlorine to produce 2 volumes of hydrogen chloride.

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$

We can see that ratio of the volumes of gaseous reactants and products is simple.

(ii) 2 volumes of carbon monooxide combine with 1 volume of oxygen to give 2 volumes of carbon dioxide $2CO + O_2 \longrightarrow 2CO_2$

(iii) One volume of nitrogen combines with 3 volumes of hydrogen to form 2 volumes of ammonia.

$N_2 + 3H_2 \longrightarrow 2NH_3$

In all above examples, we can see that ratio of volumes between the reactants and products is simple.

Thus, the law of gaseous volume is established. This law is purely a generalisation from experimental data. This law can not be proved from Dalton's atomic theory while others could be proved.

Solved Examples

Example : For the gaseous reaction $H_2 + Cl_2 \longrightarrow 2HCl$

If 40 ml of hydrogen completely reacts with chlorine then find out the required volume of chlorine and volume of produced HCl?

Solution : According to Guy Lussac's Law:

$H_2 + Cl_2 \longrightarrow 2HCl$

$\because$ 1 ml of H_2 will react will 1 ml of Cl_2 and 2 ml of HCl will be produced.

$\therefore$ 40 ml of H_2 will react with 40 ml of Cl_2 and 80 ml of HCl will produce.

Required vol. of Cl_2 = 40 ml

produced vol. of HCl = 80 ml.

Avogadro's Hypothesis

Equal volume of all gases have equal number of molecules (not atoms) at same temperature and pressure condition.

S.T.P. (Standard Temperature and Pressure)

At S.T.P. condition: temperature = 0°C or 273 K

pressure = 1 atm = 760 mm of Hg

and volume of one mole of gas at STP is found to be experimentally equal to 22.4 litres which is known as molar volume.

Mind it: Measuring the volume is equivalent to counting the number of molecules of the gas.

Solved Examples

Example : Calculate the volume in litres of 20 g hydrogen gas at STP.

Solution : No. of moles of hydrogen gas = mass ÷ atomic weight = $\frac{20\,gm}{2\,gm}$ = 10 mol

volume of hydrogen gas at STP = 10 × 22.4 lt.

PERCENTAGE COMPOSITION

Here we are going to find out the percentage of each element in the compound by knowing the molecular formula of compound.

We know that according to law of definite proportions any sample of a pure compound always possess constant ratio with their combining elements.

Solved Examples

Example : Every molecule of ammonia always has formula NH_3 irrespective of method of preparation or sources. i.e. 1 mole of ammonia always contains 1 mole of N and 3 mole of H. In other words 17 gm of NH_3 always contains 14 gm of N and 3 gm of H. Now find out % of each element in the compound.

Solution : Mass % of N in NH_3 = $\frac{\text{Mass of N in 1mol } NH_3}{\text{Mass of 1mol of } NH_3} \times 100 = \frac{14\,\text{gm}}{17\,\text{gm}} \times 100 = 82.35\ \%$

Mass % of H in NH_3 = $\frac{\text{Mass of H in 1 mol } NH_3}{\text{Mass of 1 mole of } NH_3} \times 100 = \frac{3}{17} \times 100 = 17.65\ \%$

Chemical formula

It is of two types:

(a) Molecular formulae: Chemical formulae that indicate the actual number and type of atoms in a molecule called are molecular formulae.

Example: Molecular formula of Benzene is C_6H_6

(b) Empirical formulae: Chemical formulae that indicate only the relative number of atoms of each type in a molecule are called empirical formulae.

Solved Examples

Example : Acetylene and benzene both have the empirical formula CH. The molecular masses of acetylene and benzene are 26 and 78 respectively. Deduce their molecular formulae.

Solution : ∵ Empirical Formula is CH

Step-1 The empirical formula of the compound is CH

∴ Empirical formula mass = (1 × 12) + 1 = 13.

Molecular mass = 26

Step-2 To calculate the value of 'n'

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{26}{13} = 2$$

Step-3 To calculate the molecular formula of the compound.

Molecular formula = n × (Empirical formula of the compound)

= 2 × CH = C_2H_2

Thus the molecular formula is $C_2\ H_2$

Similarly for benzene

To calculate the value of 'n'

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{78}{13} = 6$$

thus the molecular formula is 6 × CH = C_6H_6

Example : Phosgene, a poisonous gas used during World war-I, contains 12.1% C, 16.2% O and 71.7% Cl by mass. What is the empirical formula of phosgene.

(A) $COCl_2$ (B) COCl (C) $CHCl_3$ (D) $C_2O_2Cl_4$

Solution : (A)

Element	Symbol	Percentage of element	At. mass of element	Relative no. of atoms = $\frac{\text{Percentage}}{\text{At. mass}}$	Simplest atomic ratio	Simplest whole no. atomic ratio
Carbon	C	12.1	12	$\frac{12.1}{12} = 1.01$	$\frac{1.01}{1.01} = 1$	1
Oxygen	O	16.2	16	$\frac{16.2}{16} = 1.01$	$\frac{1.01}{1.01} = 1$	1
Chlorine	Cl	71.7	35.5	$\frac{71.7}{35.5} = 2.02$	$\frac{2.02}{1.01} = 2$	2

Then empirical formulae = $COCl_2$

MOLE CONCEPT

Mole: The Mass / Number Relationship

Mole is a chemical counting SI unit and defined as follows:

A mole is the amount of a substance that contains as many entities (atoms, molecules or other particles) as there are atoms in exactly 0.012 kg (or 12 gm) of the carbon-12 isotope.

From mass spectrometer we found that there are 6.023×10^{23} atoms present in 12 gm of C-12 isotope.

The number of entities in 1 mol is so important that it is given a separate name and symbol known as Avogadro constant denoted by N_A.

Mind it: In modern practice gram-atom and gram-molecule are termed as mole.

CHEMICAL EQUATION

All chemical reaction are represented by chemical equations by using chemical formula of reactants and products.

Qualitatively a chemical equation simply describes what the reactants and products are. However, a balanced chemical equation gives us a lot of quantitative information. Mainly the molar ratio in which reactants combine and the molar ratio in which products are formed.

Attributes of a balanced chemical equation:

(a) It contains an equal number of atoms of each element on both sides of equation. (POAC)

(b) It should follow law of charge conservation on either side.

(c) Physical states of all the reagents should be included in brackets.

(d) All reagents should be written in their standard molecular forms (not as atoms)

(e) The coefficients give the relative molar ratios of each reagent.

Interpretation of balanced chemical equations:

Once we get a balanced chemical equation then we can interpret a chemical equation by following ways

- Mass - mass analysis
- Mass - volume analysis
- Mole - mole analysis
- Vol - Vol analysis (separately discussed as eudiometry or gas analysis)

Now you can understand the above analysis by following example

- Mass-mass analysis:

  Consider the reaction

  $2KClO_3 \longrightarrow 2KCl + 3O_2$ According to stoichiometry of the reaction

  mass-mass ratio: 2×122.5 $2 \times 74.5 : 3 \times 32$

  or $\dfrac{\text{Mass of } KClO_3}{\text{Mass of } KCl} = \dfrac{2 \times 122.5}{2 \times 74.5}$

  $\dfrac{\text{Mass of } O_2}{\text{Mass of } KClO_3} = \dfrac{2 \times 122.5}{3 \times 32}$

- Mass - volume analysis:

  Now again consider decomposition of $KClO_3$

  $2KClO_3 \longrightarrow 2KCl + 3O_2$

  mass volume ratio:

  2×122.5 g: 2×74.5 g: 3×22.4 litre at STP we can use two relation for volume of oxygen

  $\dfrac{\text{Mass of } KClO_3}{\text{volume of } O_2 \text{at STP}} = \dfrac{2 \times 122.5}{3 \times 22.4 \text{lt}}$ ...(i)

  and $\dfrac{\text{Mass of KCl}}{\text{volume of } O_2 \text{ at STP}} = \dfrac{2 \times 74.5}{3 \times 22.4 \text{lt}}$ ...(ii)

- Mole-mole analysis:

  This analysis is very much important for quantative analysis point of view. Students are advised to clearly understand this analysis.

  Now consider again the decomposition of $KClO_3$.

  $2KClO_3 \longrightarrow 2KCl + 3O_2$

  In very first step of mole-mole analysis you should read the balanced chemical equation like 2 moles $KClO_3$ on decomposition gives you 2 moles KCl and 3 moles O_2. and from the stoichiometry of reaction we can write

  $\dfrac{\text{Moles of } KClO_3}{2} = \dfrac{\text{Moles of } KCl}{2} = \dfrac{\text{Moles of } O_2}{3}$

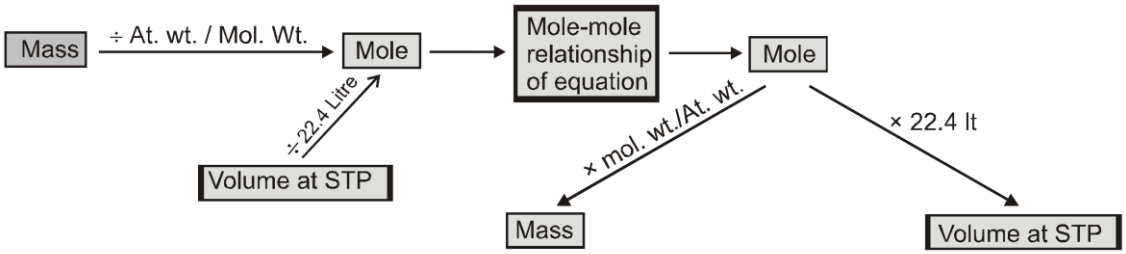
  Now for any general balance chemical equation like

  a A + b B $\longrightarrow$ c C + d D

  you can write.

  $\dfrac{\text{Moles of A reacted}}{a} = \dfrac{\text{Moles of B reacted}}{b} = \dfrac{\text{Moles of C reacted}}{c} = \dfrac{\text{Moles of D reacted}}{d}$

Mind it: In fact mass-mass and mass-vol analysis are also interpreted in terms of mole-mole analysis you can use following chart also.



Solved Examples

Example : 367.5 gram $KClO_3$ (M = 122.5) when heated how many gram of KCl and oxygen is produced.

Solution : Balance chemical equation for heating of $KClO_3$ is

$2KClO_3 \longrightarrow 2KCl + 3O_2$

mass-mass ratio:2 × 122.5 g: 2 × 74.5 g: 3 × 32 g

$$\frac{\text{mass of } KClO_3}{\text{mass of KCl}} = \frac{2\times122.5}{2\times74.5} \Rightarrow \frac{367.5}{W} = \frac{122.5}{74.5}$$

$$W = 3 \times 74.5 = 223.5 \text{ g}$$

$$\frac{\text{Mass of } KClO_2}{\text{Mass of } O_3} = \frac{2\times122.5}{3\times32} \Rightarrow \frac{367.5}{W} = \frac{2\times122.5}{3\times32} \qquad W = 144 \text{ g}$$

Example : 367.5 g $KClO_3$ (M = 122.5) when heated, how many litre of oxygen gas is produced at STP

Solution : You can use here equation (1)

$$\frac{\text{mass of } KClO_3}{\text{volume of } O_2 \text{ at STP}} = \frac{2\times122.5}{3\times22.4\text{lt}} \Rightarrow \frac{367.5}{V} = \frac{2\times122.5}{3\times22.4\text{lt}} \quad V = 3 \times 3 \times 11.2 \Rightarrow V = 100.8 \text{ lt}$$

Example : 10 ml of liquid carbon disulphide (sp. gravity 2.63) is burnt in oxygen. Find the volume of the resulting gases measured at STP.

Solution : 1 ml of CS_2 Weighs 2.63 g

10 ml of CS_2 weighs 26.3 g

CS_2	$+ 3O_2$	$\longrightarrow$	CO_2 +	$2SO_2$
12+(2×32)			22.4 L	44.8 L
76 gm	67.2 L		(total volume at STP)	

$\because$ 76 g of CS_2 will yield 67.2 L of a mixture of CO_2 and SO_2 at STP

$\therefore$ 26.3 g of CS_2 will yield $\frac{67.2}{76} \times 26.3 = 23.26$ lit.

Example : 12 L of H_2 and 11.2 L of Cl_2 are mixed and exploded. The composition by volume of mixture is–

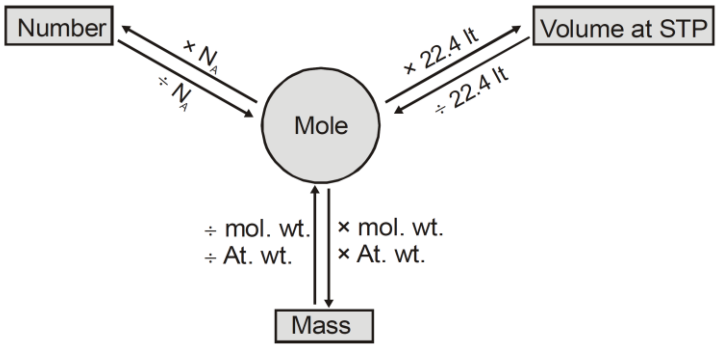
Ans. 0.8 L of H_2, 22.4 L of HCl

Solution :

H_2 +	$Cl_2 \rightarrow$	2HCl
1L	1L	2L
11.2 L	11.2 L	22.4L

Volume of H_2 = [12 – 11.2] = 0.8 L, Volume of Cl_2 = Zero, Volume of HCl = 22.4 L

Y-map: Interconversion of mole - volume, mass and number of particles:



Example : Calculate the number of gram atoms for 2×10^{23} atoms. If atomic weight of element is 24, then calculate mass of sample.

Solution : Number of gram atoms of element = $\frac{X}{N_A} = \frac{2 \times 10^{23}}{6.023 \times 10^{23}} = 0.33$ moles.

Mass of 2×10^{23} atoms = n × GAM = 0.33 × 24 = 7.92 gm

Example : Calculate total number of moles of atoms present in 49 gm H_2SO_4.

Solution : Number of mole for 49 gm H_2SO_4 = $\frac{w}{GMM} = \frac{49}{98} = 0.5$ mole

Since atomicity of $H_2SO_4 = 7$

Then total number of moles of atoms = n × atomicity = 0.5 × 7 = 3.5 moles

Example : What is the mass of 1 molecule of CO

Solution : Gram molecular weight of CO = 12 + 16 = 28 g

6.023×10^{23} molecules of CO weight 28 g

1 molecule of CO weighs = $\frac{28}{6.02 \times 10^{23}} = 4.65 \times 10^{-23}$.

Example : Calculate the volume at STP occupied by 240 gm of SO_2

Solution : Molecular weight of $SO_2 = 32 + 2 \times 16 = 64$

64 g of SO_2 occupies 22.4 litre at STP

240 g of SO_2 occupies = $\frac{22.4}{64} \times 240$ = 84 litre at STP

Example : 6×10^{20} molecules of CO_2 are removed from 220 milligram of CO_2. What are the remaining moles of CO_2

Solution : Mole of 220 mg. of CO_2 = $\frac{220 \times 10^{-3}}{44} = 5 \times 10^{-3}$ moles;

Moles of CO_2 removed are = $\frac{6 \times 10^{20}}{6 \times 10^{23}} = 10^{-3}$ moles

Remaining moles of $CO_2 = [5 \times 10^{-3} - 10^{-3}] = 4 \times 10^{-3}$ moles

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Principle of Atom Conservation (POAC):

This principle states that the moles of atoms of an element are conserved throughout the reaction.

This means, moles of an element in reactants = moles of the element in products.

Eg. $CO + O_2 \longrightarrow CO_2$

In this moles of C in CO = moles of C in CO_2

This means, moles of CO = moles of CO_2.

Similarly, for O,

moles of O in CO + moles of O in O_2 = moles of CO_2

Thus, moles of CO + 2(moles of O_2) = 2(moles of CO_2)

Advantages of Mole Method over other Methods:

(1) Balancing of chemical equations is not required in the majority of problems as the method of balancing the chemical equation is based on the principle of conservation of atom conservation.

(2) Number of reactions and their sequences, leading from reactants to products, need not be given.

Note: Whenever balanced chemical equation is given, mole method is very useful.

eg. $2KClO_3 \longrightarrow 2KCl + 3O_2$

Thus, 2(moles of $KClO_3$) = 2(moles of KCl)

3(moles of $KClO_3$) = 2(moles of O_2)

3(moles of KCl) = 2(moles of O_2)

LIMITING REAGENT

The reactant which is consumed first and limits the amount of product formed in the reaction, and is therefore, called limiting reagent.

Limiting reagent is present in least stoichiometric amount and therefore, controls amount of product. The remaining or left out reactant is called the excess reagent.

When you are dealing with balance chemical equation then if number of moles of reactants are not in the ratio of stoichiometric coefficient of balalnced chemical equation, then there should be one reactant which should be limiting reactant.

Solved Examples

Example : Three moles of Na_2CO_3 are reacted with 6 moles of HCl solution. Find the volume of CO_2 gas produced at STP. The reaction is $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

Solution : From the reaction:

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

given moles	3 mol		6 mol
given mole ratio	1	:	2
Stoichiometric coefficient ratio	1	:	2

See here given moles of reactant are in stoichiometric coefficient ratio therefore no reactant is left over.

Now use Mole-mole analysis to calculate volume of CO_2 produced at STP

$$\frac{\text{Moles of } Na_2CO_3}{1} = \frac{\text{Mole of } CO_2 \text{Produced}}{1}$$

Moles of CO_2 produced = 3

volume of CO_2 produced at STP = 3×22.4 L = 67.2 L

Example : 6 moles of Na_2CO_3 is reacted with 4 moles of HCl solution. Find the volume of CO_2 gas produced at STP. The reaction is

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

Solution : From the reaction:

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

given mole of reactant	6	:	4
give molar ratio	3	:	2
Stoichiometric coefficient ratio	1	:	2

See here given number of moles of reactants are not in stoichiometric coefficient ratio. Therefore there should be one reactant which consumed first and becomes limiting reagent.

But the question is how to find which reactant is limiting, it is not very difficult you can easily find it. According to the following method.

How to find limiting reagent:

Step: I Divide the given moles of reactant by the respective stoichiometric coefficient of that reactant.

Step: II See for which reactant this division come out to be minimum. The reactant having minimum value is limiting reagent for you.

Step: III Now once you find limiting reagent then your focus should be on limiting reagent

From Step I & II Na_2CO_3 HCl

$$\frac{6}{1}=6 \quad \frac{4}{2}=2$$

$\therefore$ HCl is limiting reagent

From Step III

From $\frac{\text{Mole of HCl}}{2}=\frac{\text{Moles of } CO_2 \text{ produced}}{1}$

$\therefore$ mole of CO_2 produced = 2 moles

$\therefore$ volume of CO_2 produced at S.T.P. = $2 \times 22.4 = 44.8$ lt.

Example : 10 moles SO_2 and 15 moles O_2 were allowed to react over a suitable catalyst. 8 moles of SO_3 were formed. The remaining moles of SO_2 and O_2 respectively are -

Ans. 2 moles, 11 moles

Solution :

$2SO_2$ +	O_2	$\longrightarrow$	$2SO_3$
10	15		8
10 – 2x	15 – x		2x
$\because$ 2x = 8	x = 4		

Hence, remaining, $SO_2 = 10 - 8 = 2$ moles, $O_2 = 15 - 4 = 11$ moles

Example : If 0.5 mol of $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4, the maximum amount of $Ba_3(PO_4)_2$ that can be formed is:

Ans. 0.10 mol

Solution :

	$3\,BaCl_2 + 2\,Na_3PO_4 \rightarrow$		$6\,NaCl$ +	$Ba_3(PO_4)_2$
molar ratio	3	2	6	1
initial moles	0.5	0.2	0	0

Limiting reagent is Na_3PO_4 hence it would be consumed, and the yield would be decided by it initial moles.

2 moles of Na_3PO_4 give 1 mole of $Ba_3(PO_4)_2$,

$\therefore$ 0.2 moles of Na_3PO_4 would give 0.1 mole of of $Ba_3(PO_4)_2$

EUDIOMETRY OR GAS ANALYSIS:

Eudiometry means gas analysis. It means in this type of numericals either you have to find out vol. of gas from the data given in question or you have to find out any other thing like molecular formula of hydrocarbon ,no. of mole , mass of any other reactant and vol. of gas is given. Specific gases are absorbed in specific reagents as given below:

Reactants	**Absorbed gases**
KOH solution	CO_2 & SO_2
Alkaline Pyrogallol	O_2
Solution of ammonical cuprous chloride	CO

And if the mixture is cooled, H_2O vapour changes to liquid.

Hence, when temperature is above 100°C (373 K) then only volume of H_2O is considered.

Eudiometry is mainly based on Avogadro's law,

$$A(g) + B(g) \rightarrow C(g) + D(g)$$

a volume b volume c volume d volume

a moles b moles c moles d moles

Solved Examples

Example : Three moles of Na_2CO_3 is reacted with 6 moles of HCl solution. Find out the volume of CO_2 gas produced at STP

Solution : $Na_2CO_3 \rightarrow NaCl + CO_2$

Apply POAC on C atom

No. of moles of C atoms in Na_2CO_3 = no. of moles of C atoms in CO_2

1* no. of moles of Na_2CO_3 = 1* no. of moles of CO_2

So no. of moles of Na2CO3 = no. of moles of CO_2 = 3 moles (given)

Vol. of CO_2 = 3 * 22.4 litre = 67.2 litre

Example : 10 ml. of a gaseous hydrocarbon was burnt completely in 80 ml. of O_2 at NTP. The remaining gas occupied 70 ml. at NTP. This vol. became 50 ml on treatment with KOH solution. What is the molecular formula of hydrocarbon?

Solution : $C_x H_y + O_2 \rightarrow CO_2 + H_2O$

Vol. of (CO_2 + unreacted O_2) = 70 ml.

Because CO_2 can be absorbed by KOH. So when this 70 ml. mix. Is treated it becomes 50 ml it means vol. of CO_2 in the mixture was 20 ml. which is absorbed by KOH and remaining 50 ml. was unreacted O_2 (O_2 can NOT absorbed by KOH)

So vol. of CO_2 = 20 ml.

Vol. of O_2 reacted = 80 – 50 = 30 ml.

Vol. of hydrocarbon = 10 ml.

Apply POAC on C atoms

X × no. of mole of $C_x H_y$ = 1× no. of mole of CO_2 …(i)

X× 10 = 20

So X = 20/10= 2

Apply POAC on H atoms

Y× no. of moles of $C_x H_y$ = 2× no. of mole of H_2O …(ii)

Apply POAC on O atoms

2× no. of moles of O_2 = 1× no. of mole of H_2O + 2× no. of mole of CO_2

From eq. (ii)

2× no. of moles of O_2 = y/2× no. of mole of $C_x H_y$ + 2× no. of mole of CO_2

2×30 = Y/2×10 + 2 ×20

20 = 10 Y/2

Y = 4

Put the value of X and Y in $C_x H_y$

The molecular formula of hydrocarbon is C_2H_4

CONCENTRATION TERMS

A mixture of two or more substances can be a solution. We can also say that a solution is a homogeneous mixture of two or more substances. 'Homogeneous' means 'uniform throughout'. Thus a homogeneous mixture, i.e., a solution, will have uniform composition throughout.

Properties of a solution:

- A solution is clear and transparent. For example, a solution of sodium chloride in water is clear and transparent.
- The solute in a solution does not settle down even after the solution is kept undisturbed for some time.
- In a solution, the solute particle cannot be distinguished from the solvent particles or molecules even under a microscope. In a true solution, the particles of the solute disappear into the space between the solvent molecules.
- The components of a solution cannot be separated by filtration.

  The following concentration terms are used to expressed the concentration of a solution. These are
- Molarity (M) • Molality (m) • Mole fraction (x) • % calculation • ppm
- Remember that all of these concentration terms are related to one another. By knowing one concentration term you can also find the other concentration terms. Let us discuss all of them one by one.

(a) Molarity (M)

The number of moles of a solute dissolved in 1 L (1000 ml) of the solution is known as the molarity of the solution.

i.e., Molarity of solution = $\frac{\text{volume of solution in litre}}{\text{number of solution in litre}}$

number of moles

Let a solution is prepared by dissolving w gm of solute of mol. wt. M in V ml water.

$\therefore$ Number of moles of solute dissolved = $\frac{w}{M}$

$\therefore$ V ml water have $\frac{w}{M}$ mole of solute

$\therefore$ 1000 ml water have $\frac{w \times 1000}{M \times V_{(ml)}}$ $\quad\therefore$ Molarity (M) = $\frac{w \times 1000}{(\text{Mol. wt of solute}) \times V_{(ml)}}$

Some other relations may also be useful.

Number of millimoles = $\frac{\text{mass of solute}}{(\text{Mol. wt of solute})} \times 1000$ = (Molarity of solution × V_{ml})

- **Molarity of solution may also given as:**

$$\frac{\text{Number of millimole of solute}}{\text{Total volume of solution in ml}}$$

- Molarity is an unit that depends upon temperature. It varies inversely with temperature.

Mathematically: Molarity decreases as temperature increases.

$$\text{Molarity} \propto \frac{1}{\text{temperature}} \propto \frac{1}{\text{volume}}$$

- If a particular solution having volume V_1 and molarity = M_1 is diluted to V_2 mL the

$M_1V_1 = M_2V_2$

M_2: Resultant molarity

- If a solution having volume V_1 and molarity M_1 is mixed with another solution of same solute having volume V_2 & molarity M_2

then $M_1V_1 + M_2V_2 = MR\,(V_1 + V_2)$

MR = Resultant molarity = $\frac{M_1V_1 + M_2V_2}{V_1 + V_2}$

Solved Examples

Example : 149 gm of potassium chloride (KCl) is dissolved in 10 Lt of an aqueous solution. Determine the molarity of the solution (K = 39, Cl = 35.5)

Solution : Molecular mass of KCl = 39 + 35.5 = 74.5 gm

$\therefore$ Moles of KCl = $\frac{74.5\text{ gm}}{149\text{ gm}} = 2$

$\therefore$ Molarity of the solution = $\frac{2}{10} = 0.2$ M

(b) Molality (m):

The number of moles of solute dissolved in1000 g (1 kg) of a solvent is known as the molality of the solution.

i.e., molality = $\frac{\text{number of moles of solute}}{\text{mass of solvent in gram}} \times 1000$

Let y g of a solute is dissolved in x g of a solvent. The molecular mass of the solute is m. Then Y/m mole of the solute are dissolved in x g of the solvent. Hence

Molality = $\frac{Y}{m \times x} \times 1000$

- Molality is independent of temperature changes.

Relation Between Molality And Molarity:

$$\frac{\text{Molarity}}{\text{Molality}} = \frac{\text{Moles of solute}}{\text{volume of solution (in litre)}} \times \frac{\text{mass of solvent(kg)}}{\text{moles of solute}} = \frac{\text{mass of solvent(kg)}}{\text{volume of solution (in litre)}}$$

$$= \frac{\text{mass of solvent(in grams)}}{\text{volume of solution (in mL)}} = \frac{W}{V}$$

Let the density of the solution be d g mL^{-1}

Mass of solution = V × d

Mass of solute = number of moles × molecular mass of solute

= n × m_A

Mass of solvent(W) = mass of solution – Mass of solute

= (V × d) – (n × m_A)

Thus, $\frac{\text{Molarity}}{\text{Molality}} = \frac{(V\times d)-(n\times m_A)}{V}$

$\text{Molality (m)} = \frac{\text{Molarity}\times V}{(V\times d)-(n\times m_A)}$

$m = \frac{\text{Molarity}}{d-\left(\frac{n}{V}\times m_A\right)} = \frac{\text{Molarity}\times 1000}{d-(\text{Molarity}\times m_A)}$

Solved Examples

Example : 255 g of an aqueous solution contains 5 g of urea. What is the concentration of the solution in terms of molality. (Mol. wt. of urea = 60)

Solution : Mass of urea = 5 g

Molecular mass of urea = 60

Number of moles of urea = $\frac{5}{60} = 0.083$

Mass of solvent = (255 – 5) = 250 g

∴ Molality of the solution

$= \frac{\text{Number of moles of solute}}{\text{Mass of solvent in gram}}\times 1000 = \frac{0.083}{250}\times 1000 = 0.332$

(c) Mole fraction:

The ratio of number of moles of the solute or solvent present in the solution and the total number of moles present in the solution is known as the mole fraction of substances concerned.

Let number of moles of solute in solution = n

Number of moles of solvent in solution = N

∴ Mole fraction of Solute (x_1) = $\frac{n}{n+N}$

∴ Mole fraction of solvent (x_2) = $\frac{N}{n+N}$

also $x_1 + x_2 = 1$

- Mole fraction is a pure number. It will remain independent of temperature changes.

Solved Examples

Example : 46g C_2H_5OH and 54g H_2O are mixed together. Calculate mole fraction of each.

Solution : Moles of $C_2H_5OH = \frac{46}{46} = 1$ mole

Moles of $H_2O = \frac{54}{18} = 3$ mole

$$X_{C_2H_5OH} = \frac{1}{1+3} = 0.25$$

$$X_{H_2O} = \frac{3}{1+3} = 0.75$$

(d) % calculation:

The concentration of a solution may also expressed in terms of percentage in the following way.

- % weight by weight (w/w): It is given as mass of solute present per 100 g of solution.

  i.e. % w/w = $\frac{\text{mass of solute in g}}{\text{mass of solution in g}} \times 100$

- % weight by volume (w/v): It is given as mass of solute present per 100 ml of solution.

  i.e., % w/v = $\frac{\text{mass of solute in g}}{\text{Volume of solution in ml}} \times 100$

- % volume by volume (V/V): It is given as volume of solute present per 100 ml solution.

  i.e., % V/V

  = $\frac{\text{Volume of solute in mL}}{\text{Volume of solution in mL}} \times 100$

Solved Examples

Example : 0.5 g of a substance is dissolved in 25 g of a solvent. Calculate the percentage amount of the substance in the solution.

Solution : Mass of substance = 0.5 g

Mass of solvent = 25 g

$\therefore$ percentage of the substance (w/w) = $\frac{0.5}{0.5+25} \times 100 = 1.96.$

Example : 20 cm^3 of an alcohol is dissolved in 80 cm^3 of water. Calculate the percentage of alcohol in solution.

Solution : Volume of alcohol = 20 cm^3

Volume of water = 80 cm^3

$\therefore$ percentage of alcohol = $\frac{20}{20+80} \times 100 = 20$

Example : What is the concentration of sugar ($C_{12}H_{22}O_{11}$) in mole L^{-1} if its 20g are dissolved in enough water to make a final volume upto 2L?

Solution : Molarity of solution (mol L^{-1}) = V in ML $\frac{\text{mass of solute(g)}}{\text{M.Mass}} \times \frac{1000}{\text{V in ML}}$

Conc. of sugar = $\frac{20}{342} \times \frac{1000}{2000}$ = 0.0292 mol L^{-1}

Example : If the density of methanol is 0.793 kg L^{-1}, what is its volume needed for making 2.5 L of its 0.25 M solution?

Solution : Amount of methanol required m(g) = $\frac{\text{Molarity} \times \text{M.Mass} \times V_{in}\text{ML}}{1000} = \frac{0.25 \times 32 \times 2500}{1000} = 20\text{g}$

Since, density = $\frac{\text{Mass}}{\text{Volume}}$;

$\therefore$ Volume of CH_3 OH required = density $\frac{\text{Mass}}{\text{Density}} = \frac{20}{0.793} = 25.22$ mL

Example : A given solution of NaOH contains 2.00 g of NaOH per litre of solution. Calculate the molarity of this solution.

Solution : Molar mass of NaOH = 40.00 g

2.00 g NaOH = $\frac{2.00}{40.00}$ mole NaOH = 0.05 mole NaOH

Molarity = $\frac{\text{Mole of solute}}{\text{Volume of solution in litres}} = \frac{0.05}{1} = 0.05$ M

Example : 2.8 g of KOH is dissolved in water to give 200 cm^3 of solution. Calculate the molarity of KOH in the solution.

Solution : Molar mass of KOH = 56.00 g

2.8 g of KOH = $\frac{2.8}{56}$ moles of KOH = 0.05 mole of KOH

Molarity = $\frac{\text{Mole of solute}}{\text{Volume of solution in litres}} = \frac{0.05}{200/1000} = 0.25$ M

Example : Calculate the number of moles and the amount in gram of NaOH in 250 cm^3 of a 0.100 M NaOH solution.

Solution : A 0.100 M NaOH solution contains 0.1 moles of NaOH in 1 L or 1000 cm3 of solution.

Number of moles of NaOH in 250 cm^3 of solution = $\frac{0.1 \times 250}{1000} = 0.025$ mole NaOH

Molar mass of NaOH = 40.00 g

Mass of NaOH in 250 cm^3 of solution = 0.025 × 40.00 = 1.0 g

Example : How much volume of 10 M HCl should be diluted with water to prepare 2.00 L of 5 M HCl?

Solution : In dilution the following equation is applicable:

$M_1 V_1$	=	$M_2 V_2$
10 M HCl		5 M HCl

$10 \times V_1 = 5 \times 2.00$

$V_1 = \frac{5 \times 2.00}{10}$ L = 1.00 L

Example : What volume of 6 M HCl and 2 M HCl should be mixed to get one litre of 3 M HCl?

Solution : Suppose the volume of 6 M HCl required to obtain 1 L of 3 M HCl = x L

$\therefore$ Volume of 2 M HCl required = (1 – x) L

Applying the molarity equation

$M_1 V_1$ +	$M_2 V_2$	=	$M_3 V_3$
6 M HCl	2 M HCl		3 M HCl

$$6x + 2(1 - x) = 3 \times 1$$

$$6x + 2 - 2x = 3$$

$$4x = 1$$

$$x = 0.25 \text{ L}$$

Hence volume of 6 M HCl required = 0.25 L

Volume of 2 M HCl required = 0.75 L

Example : **(a)** A sample of NaOH weighing 38 g is dissolved in water and the solution is made to 50.0 mL in a volumetric flask. What is the molarity of the resulting solutions?

(b) How many moles of NaOH are contained in 27 mL of 0.15 M NaOH ?

Solution : **(a)** Moles of NaOH in 50.0 mL solution = $\frac{38}{40}$

$\therefore$ moles of NaOH in 1.0 L solution = $\frac{38}{40} \times \frac{1000}{50} = 19$ M

(b) 1.0 L of solution contains moles = 0.15

$\therefore$ 27.0 mL of solution contains moles = $\frac{0.15}{1000} \times 27 = 4.05 \times 10^{-3}$ moles.

Example: Calculate the mass of sodium acetate (CH_3COO Na) required to make 500 mL of 0.375 molar aqueous solutions. Molar mass of sodium acetate is 82.0245 g $mole^{-1}$.

Solution : Molarity of solution (M) = $\frac{\text{mass of solute (g)}}{\text{M. Mass}} \times \frac{1000}{\text{V in mL}}$

Hence, mass of solute = $\frac{\text{Molarity} \times \text{M.Mass} \times \text{Volume of solution}}{1000}$

$\therefore$ Mass of sodium acetate required = $\frac{0.375 \times 82.0245 \times 500}{1000} = 15.379 = 15.38$ g

EXERCISE #1

DALTON'S ATOMIC THEORY

1. Which postulate of Dalton's Atomic theory suggests the law of conservation of mass?

(A) Atoms cannot be divided.

(B) All the matter is made up of very small particles called atoms.

(C) Elements consists of atoms combined in a fixed ratio.

(D) Atoms can neither be created nor be destroyed.

2. Dalton incorporated the law of conservation of mass into his atomic theory by asserting that-

(A) Matter is composed of atom

(B) Atom can be destroyed in the chemical reactions.

(C) Atoms are indivisible.

(D) Atoms can be combined in any ratio of their mass.

3. Dalton's Atomic theory agrees with modern atomic theory except for the statement that-

(A) Atoms are not divided in chemical reactions.

(B) all atoms of same elements have same mass.

(C) Atoms of the same element are chemically alike.

(D) Atoms can be combined in any ratio of their mass.

LAWS OF CHEMICAL COMBINATION

1. All samples of carbon dioxide contain carbon and oxygen in the mass ratio 3: 8. This is in agreement with the law of

(A) conservation of mass (B) constant proportions

(C) (A) and (B) both (D) none of these

2. A chemical equation is always balanced to fulfil the condition of -

(A) Law of conservation of mass. (B) Law of multiple proportions.

(C) Law of constant proportions (D) All of these

3. The law of conservation of mass was given by

(A) John Dalton (B) Proust (C) Lavoisier (D) None of these

4. If the mass of reactants is equal to the mass of products then which of the following statements is true?

(A) Law of conservation of mass holds good.

(B) Mass can neither be created nor be destroyed.

(C) There is no change in mass during a chemical reaction.

(D) All of the above

5. Hydrogen & Oxygen combine in the ratio of 1: 8 by mass to form water. What mass of oxygen gas would be required to react completely with 3 g of hydrogen gas?

(A) 24 g (B) 27 g (C) 21 g (D) 3 g

6. Ammonia contains 82.65% N_2 and 17.65% H_2. If law of constant proportion is true, then mass of zinc required to give 10g Ammonia will be:

(A) 8.265 g (B) 0.826 g (C) 82.65 g (D) 82.65 g

7. Irrespective of source, pure H_2O sample of H_2O always yields 88.89% mass of oxygen and 11.11% mass of hydrogen. This is explained by law of -

(A) conversation of mass (B) constant composition

(C) multiple proportion (D) constant volume

8. One part of an element A combines with 2 parts of b another element 6 parts of element C combine with 4 parts of element B. If A & C combine together, the ratio of their masses will be governed by:

(A) law of definite proportions (B) law of multiple proportions

(C) law of reciprocal proportions (D) law of conservation of mass.

9. How much volume of oxygen will be required for complete combustions of 40ml of acetylene and how much volume of carbondioxide will be formed. All volume we measured at NTP.

(A) 100, 50 (B) 100, 60 (C) 100, 80 (D) 100, 40

10. What would be the mass of water formed by reaction of 2 grams hydrogen with 16g oxygen?

(A) 16 g (B) 32 g (C) 18 g (D) 20 g

11. Hydrogen combines with chlorine to form HCl. It also combines with sodium to form NaH. If sodium and chlorine also combine with each other, they will do so in ratio of masses.

(A) 23:35.5 (B) 35.5:23 (C) 1:1 (D) 23:1

12. The law of multiple proportions is illustrated by pair of compounds:

(A) NaCl, NaBr (B) H_2O, D_2O (C) SO_2, SO_3 (D) HgOH, HgO

13. A sample of $CaCO_3$ has following % composition: Ca = 40%, C = 12%, O = 48%. If law of constant proportions is true, then weight of Ca in 4g of a sample of calcium in 4g of a sample of $CaCO_3$ obtained from another source will be.

(A) 0.016 g (B) 0.16 g (C) 1.6 g (D) 16 g

14. 4.8 g of $KClO_3$ when heat produced 1.9 g of oxygen and the residue (KCl) left behind 2.9 g. Which law of chemical combination is illustrated?

(A) law of constant proportion (B) law of multiple proportion

(C) law of conservation of mass. (D) law of reciprocal proportion

15. H_2O_2 and H_2O contains 5.93% and 11.2% of H_2 respectively. Which law of chemical combination is:

(A) law of constant proportion

(B) law of multiple proportion

(C) law of conservation of mass.

(D) law of reciprocal proportion

PERCENTAGE COMPOSITION

1. The volume percentage of the solute in 250 ml of a solution containing 5gm of sugar is

(A) 4% (B) 2% (C) 5% (D) 9%

2. The amount of glucose required to prepare a 250gm of 5% solution of glucose by mass is

(A) 12.5g (B) 15g (C) 17g (D) 20g

3. Calculate the percentage composition of calcium in calcium nitrate.

(A) 24% (B) 12% (C) 55% (D) 9%

4. Calculate the percentage composition of water of crystalisation in $FeSO_4.7H_2O$

(A) 45.3% (B) 52% (C) 25% (D) 19%

5. It is found that 16.5 g of metal combine with oxygen to form 35.60 g of metal oxide. Calculate the percentage of metal and oxygen in the compound.

(A) 46, 54% (B) 20, 80% (C) 55, 45% (D) 25, 75%

6. Density of dry air containing only N_2 and O_2 is 1.15g/L at 740 mm and 300K. What is % composition of N_2 by weight in the air?

(A) 78% (B) 75.5% (C) 70.02% (D) 73.42%

7. A sample of calcium carbonate ($CaCO_3$) has the following percentage composition: Ca = 40%, C = 12%, O = 48%. If the law of constant proportions is true, then the weight of calcium in 4g of a sample of calcium carbonate obtained from another source will be:

(A) 0.016 g (B) 0.16 g (C) 1.6 g (D) 16 g

8. What percentage of oxygen is present in the compound $CaCO_3.3Ca_3(PO_4)_2$?

(A) 23.3% (B) 45.36% (C) 41.94% (D) 17.08%

9. 2.0 g sample contain mixture of SiO_2 and Fe_2O_3, on very strong heating leave a residue weighing 1.96 g. The reaction responsible for loss of weight is

Fe_2O_3 (s) $\longrightarrow$ Fe_3O_4(s) + O_2(g), (unbalance equation)

What is the percentage by mass of SiO_2 in original sample?

(A) 10% (B) 20% (C) 40% (D) 60%

10. The weight of a mixture containing HCl and H_2SO_4 is 0.1g. On treatment with an excess of an $AgNO_3$ solution, reacted with this acid mixture gives 0.1435g of AgCl. Weight % of the H_2SO_4 is mixture is:

(A) 36.5 (B) 63.5 (C) 50 (D) None of these

MOLE CONCEPT

1. Which of the following has the smallest mass?

(A) 4 g of He (B) 6.023×10^{23} atoms of He

(C) 1 atom of He (D) 1 mole atoms of He

2. The number of carbon atoms in 1 g of $CaCO_3$ is -

(A) 6.023×10^{23} (B) 6.023×10^{21} (C) 3.0125×10^{22} (D) 1.204×10^{23}

3. 6.023×10^{20} atoms of silver (Atomic mass = 108 u) weigh -

(A) 108×10^3 g (B) 108 g (C) 0.108 g (D) 10.8 g

4. Which of the following has largest number of molecules?

(A) 8 g of CH_4 (B) 4.4 g of CO_2

(C) 34.2 g of $C_{12}H_{22}O_{11}$ (D) 2 g of H_2

5. Which of the following contains one mole molecules of the substance?

(A) 16 g Oxygen (B) 7 g Nitrogen (C) 2 g Hydrogen (D) 36 g Water.

6. The number of molecules in 16.0 g of oxygen is -

(A) 6.02×10^{23} (B) 6.02×10^{-23} (C) 3.01×10^{-23} (D) 3.01×10^{23}

7. The volume of one mole of a gas at normal temperature and pressure is -

(A) 11.2 litres (B) 22.4 litres (C) 100 litres (D) None of these

8. The number of gram atoms in 8 g of He are -

(A) 2 (B) 1.204×10^{24} (C) 3.1×10^{23} (D) None of these

9. The percentage of hydrogen in H_2O is -

(A) 8.88 (B) 11.12 (C) 20.60 (D) 80.0

10. The charge in coulombs of 1 gram ion of N^{3-} is (the charge on an electron is 1.602×10^{-19} C)-

(A) 2.894×10^5 C (B) 3.894×10^5 C (C) 2.894×10^6 C (D) None of these

11. The mass of oxygen contained in 1 kg of potassium nitrate (KNO_3) will be -

(A) 478.5 g (B) 485.5 g (C) 475.2 g (D) 488. 2 g

12. 1 mole of a compound contains 1 mole of carbon and 2 moles of oxygen. The molecular weight of the compound is -

(A) 3 (B) 12 (C) 32 (D) 44

13. What amount of CaO will be produced by 1 g of calcium?

(A) 2.4 g (B) 1.4 g (C) 2.1 g (D) 3 g

14. A sample of $KClO_3$ on decomposition yielded 448 mL of oxygen gas at NTP. Calculate the moles of oxygen produced:

(A) 0.02 (B) 0.03 (C) 0.2 (D) 2

15. In Haber process 30 litres of dihydrogen and 30 litres of dinitrogen were taken for reaction which yielded only 50% of the expected product. What will be the volume of dinitrogen and dihydrogen consumed during the reaction?

(A) H_2 = 10L, N_2 = 5L (B) N_2 = 30L, H_2 = 5L (C) N_2 = 15L, H_2 = 5L (D) N_2 = 10L, H_2 = 50L

16. The weight of oxygen required for the complete combustion of 2.8 g of ethylene is

(A) 9.6 g (B) 1.4 g (C) 2.1 g (D) 3 g

17. Which of the following sample contains the maximum number of atoms -

(A) 1 mg of C_4H_{10} (B) 1 mg of N_2 (C) 1 mg of Na (D) 1 mL of water

18. The total number of protons, electrons and neutrons in 12 g of ${}_6C^{12}$ is -

(A) 1.084×10^{25} (B) 6.022×10^{23} (C) 6.022×10^{22} (D) 18

19. 4.4 g of CO_2 and 2.24 litre of H_2 at STP are mixed in a container. The total number of molecules present in the container will be -

(A) 6.022×10^{23} (B) 1.2044×10^{23} (C) 2 mole (D) 6.023×10^{24}

20. How many atoms are contained in a mole of $Ca(OH)_2$ -

(A) $30 \times 6.02 \times 10^{23}$ atoms/mol (B) $5 \times 6.02 \times 10^{23}$ atoms/mol

(C) $6 \times 6.02 \times 10^{23}$ atoms/mol (D) None of these

21. Which of the following has the maximum mass?

(A) 25 g of Hg (B) 2 moles of H_2O (C) 2 moles of CO_2 (D) 4 g atom of oxygen

22. If the atomic mass of Sodium is 23, the number of moles in 46 g of sodium is:

(A) 1 (B) 2 (C) 2.3 (D) 4.6

23. Number of moles present in 1 m^3 of a gas at NTP are -

(A) 44.6 (B) 40.6 (C) 42.6 (D) 48.6

24. The volume occupied by one molecule of water (density 1 g/cm^3) is -

(A) 18 cm^3 (B) 22400 cm^3 (C) 6.023×10^{-23} (D) 3.0×10^{-23} cm^3

25. Total mass of neutrons in 7mg of ${}_{14}C$ is -

(A) 3×10^{20} kg (B) 4×10^{-6} kg (C) 5×10^{-7} kg (D) 4×10^{-7} kg

LIMITING REAGENT

1. If 0.5 mol of $BaCl_2$ is mixed with 0.1 mole of Na_3PO_4, the maximum number of mole of $Ba_3(PO_4)_2$ that can be formed is:

(A) 0.7 (B) 0.05 (C) 0.30 (D) 0.10

2. For the reaction $2P + Q \rightarrow R$, 8 mol of P and 5 mol of Q will produce

(A) 8 mol of R (B) 5 mol of R (C) 4 mol of R (D) 13 mol of R

3. 4 g of hydrogen is ignited with 4 g of oxygen, the amount of water formed is?

(A) 2.5 g (B) 0.5 g (C) 4.5 g (D) 8 g

4. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of moles of $CaSO_4$ formed is:

(A) 0.2 (B) 0.5 (C) 0.4 (D) 1.5

5. For the reaction: $A + 2B \rightarrow C$
5 mole of A and 8 mole of B will produce:
(A) 5 mole of C (B) 4 mole of C (C) 8 mole of C (D) 12 mole of C

6. Zinc and hydrochloric acid react according to the reaction:
$Zn(s) + 2HCl(aq.) \longrightarrow ZnCl_2(aq.) + H_2(g)$
If 0.3 mole of Zn are added to hydrochloric acid containing 0.52 mole HCl, how many moles of H_2 are produced:
(A) 0.26 (B) 0.52 (C) 0.14 (D) 0.30

7. In Haber process, 30 litres of dihydrogen and 30 litres of dinitrogen were taken for reaction which yielded only 50% of the expected product. What will be the composition of the gaseous mixture under the aforesaid condition in the end?
(A) 20 litres NH_3, 25 litres N_2, 20 litres H_2 (B) 10 litres NH_3, 25 litres N_2, 15 litres H_2
(C) 20 litres NH_3, 10 litres N_2, 30 litres H_2 (D) 20 litres NH_3, 25 litres N_2, 15 litres H_2

CONCENTRATION TERMS

1. 100 mL of 1 M HCl is mixed with 50 mL of 2 M HCl. Hence, $[H_3O^+]$ is:
(A) 1.00 M (B) 1.50 M (C) 1.33 M (D) 3.00 M

2. The mass percentage of sugar in a solution containing 47g of sugar and 72g of water is
(A) 40% (B) 30% (C) 11% (D) 7%

3. The mass percentage of sugar in a solution containing 100g of sugar and 100g of water is
(A) 40% (B) 50% (C) 11% (D) 7%

4. 4.9 g of H_2SO_4 is present in 100 mL of the solution. What is the molarity of the solution?
(A) 0.2M (B) 0.5M (C) 0.21M (D) 0.25M

5. What volume of a 0.8 M solution contains 100 millimoles of the solute?
(A) 80 mL (B) 125 mL (C) 125 L (D) 80 L

6. Calculate the molarity when:
(a) 4.9 g H_2SO_4 acid dissolved in water to result 500 mL solution.
(b) 2 gram-molecules of KOH dissolved in water to result 500 mL solution.
(A) (a) 0.1 M (b) 0.07 M (B) (a) 0.4 M (b) 4 M
(C) (a) 0.4 M (b) 0.07 M (D) (a) 0.1 M (b) 4 M

7. Calculate the amount of 75% pure NaI required to prepare 5 litre of 0.5 M solution.
(A) 281.25 g (B) 500 g (C) 923.33 g (D) 519.375 g

8. Concentrated HNO_3 is 63% HNO_3 by mass and has a density of 1.4 g/mL. How many millilitres of this solution are required to prepare 250 mL of a 1.20 M HNO_3 solution?
(A) 18.0 (B) 21.42 (C) 20.0 (D) 14.21

9. What value of 75% alcohol by weight (d = 0.80 g/cm3) must be used to prepare 150 cm3 of 30% alcohol by weight (d = 0.90 g/cm3)?
(A) 67.5 mL (B) 56.25 mL (C) 44.44 mL (D) None of these

10. H_3PO_4 (98 g mol^{-1}) is 98% by mass of solution. If the density is 1.8 g/ml, the molarity is

(A) 18 M (B) 36 M (C) 54 M (D) 0.18 M

11. How would you prepare exactly 3 litre of 1 M NaOH solution by mixing proportions of stock solutions of 2.5 M NaOH and 0.4 M NaOH, if no water is to be used. Find the ratio of the volume (V_1/V_2).

(A) 1: 3 (B) 3: 7 (C) 2: 5 (D) Data insufficient

12. What approximate volume of 0.40 M $Ba(OH)_2$ solution must be added to 50.0 mL of 0.30 M the molarity of the OH^- ions is 0.50 M ?

(A) 33 mL (B) 66 mL (C) 133 mL (D) 100 mL

13. 49 gram of H_2SO_4 is present in 67 g of solution mole fraction of water will be:

(A) 0.33 (B) 0.66 (C) 0.6 (D) 0.5

14. 50 g of 40% w/w NaOH contain. Calculate mole fraction of water?

(A) 0.23 (B) 0.76 (C) 0.3 (D) 0.7

15. 100 ml of 80% w/v NaOH contain. Calculate mole fraction of NaOH?

(A) 0.64 (B) 0.36 (C) 0.23 (D) 0.76

16. 100 g of water is added to 0.585 g of NaCl what is the molarity of $Cl^{\ominus}$ ion?

(A) 10^{-4} M (B) 10^{-1} M (C) 10^{-2} M (D) 10^{-3} M

17. In 2 mole of solution mole fraction of water is 0.8. Moles of solute will be?

(A) 0.4 (B) 1.6 (C) 0.8 (D) 1.2

18. 40.25 g of Glauber's salt is dissolved in water to obtain 500 mL of solution of density 1077.5 g dm^{-3}. The molality of Na_2SO_4 in solution is about

(A) 0.35 mol kg^{-1} (B) 0.24 mol kg^{-1} (C) 0.46 mol kg^{-1} (D) 0.17 mol kg^{-1}

19. The mole fraction of ethanol in water is 0.08. Its molality will be

(A) 2.41 mol kg^{-1} (B) 4.83 mol kg^{-1} (C) 3.33 mol kg^{-1} (D) 6.41 mol kg^{-1}

EXERCISE #2

1. A 20.00 mL sample of a $Ba(OH)_2$ solution is titrated with 0.245 M HCl. If 27.15 mL of HCl is required, what is the molarity of the $Ba(OH)_2$ solution?

(A) 0.166 M (B) 0.180 M (C) 0.333 M (D) 0.666 M

2. How many moles of Na^+ ions are in 20 mL of 0.40 M Na_3PO_4?

	Molar Mass, M
$Al_2(SO_4)_3 \cdot 18H_2O$	666.43 g·mol

(A) 0.0080 (B) 0.024 (C) 0.050 (D) 0.20

3. What is the mass percent of oxygen in $Al_2(SO_4)_3 \cdot 18H_2O$?

(A) 9.60 (B) 28.8 (C) 43.2 (D) 72.0

4. What is the number of O_2 molecules in the 2.5 g of O_2 inhaled by the average person in one minute?

(A) 1.9×10^{22} (B) 3.8×10^{22} (C) 4.7×10^{22} (D) 9.4×10^{22}

5. If 1.50 g of $H_2C_2O_4.2H_2O$ were heated to drive off the water of hydration, how much anhydrous $H_2C_2O_4$ would remain?

(A) 0.34 g (B) 0.92 g (C) 1.07 g (D) 1.50 g

6. How many mL of 8.00 M HCl are needed to prepare 150. mL of a 1.60 M HCl solution?

(A) 30.0 mL (B) 24.0 mL (C) 18.8 mL (D) 12.0 mL

7. Analysis of a compound known to contain only Mg, P, and O gives this analysis. 21.8% Mg 27.7% P 50.3% O. What is its empirical formula?

(A) $MgPO_2$ (B) $MgPO_3$ (C) $Mg_2P_2O_7$ (D) $Mg_3P_2O_8$

8. The reaction of ethanol, C_2H_5OH, with oxygen is a popular classroom demonstration. Balance the equation to find the number of moles of gaseous products formed per mole of ethanol.

$___C_2H_5OH(g) + ___O_2(g) \rightarrow ___CO_2(g) + ___H_2O(g)$

(A) 2 (B) 3 (C) 4 (D) 5

9. Ammonia is produced in accordance with this equation.

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

In a particular experiment, 0.25 mol of NH_3 is formed when 0.5 mol of N_2 is reacted with 0.5 mol of H_2. What is the percent yield?

(A) 75% (B) 50% (C) 33% (D) 25%

10. A chloride salt of rhenium contains 63.6% Re by mass. What is its empirical formula?

(A) ReCl (B) $ReCl_2$ (C) $ReCl_3$ (D) $ReCl_5$

11. The amount of chloride ion in a water sample is to be determined by adding excess silver nitrate. If 1.0 g of silver chloride is precipitated, what mass of chloride ion is in the original sample?

(A) 0.25 g (B) 0.34 g (C) 0.50 g (D) 0.75 g

12. How many millimoles of methane, CH_4, are present in 6.4 g of this gas?

(A) 0.40 (B) 4.0 (C) 40 (D) 4.0×10^2

13. According to the equation,

$SnO_2 + 2H_2 \rightarrow Sn + 2H_2O$

What volume of hydrogen, measured at 1 atm and 273 K, is required to react with 2.00 g of SnO_2?

(A) 0.00133 L (B) 0.00265 L (C) 0.297 L (D) 0.595 L

14. An oxide of manganese contains 2.29 g of manganese per gram of oxygen. What is the empirical formula of this compound?

(A) MnO (B) MnO_2 (C) Mn_2O_3 (D) MnO_3

15. A heterogeneous system is produced when 0.040 moles of solid NaCl is added to 0.10 L of 0.10 M $Pb(NO_3)_2$. Which ion is present in the aqueous phase at the highest concentration?

(A) Cl^- (B) NO_3^- (C) Pb^{2+} (D) Na^+

16. Which expression gives the fraction by mass of nitrogen in ammonium dihydrogen phosphate?

(A) 14 / 115 (B) 28 / 115 (C) 28 / 132 (D) 14 / 210

17. Ethanol burns in excess oxygen to form $CO_2(g)$ and $H_2O(g)$ according to this balanced equation.

$C_2H_5OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

What value is closest to the volume of $CO_2(g)$, measured at 200K and 1 atm, produced from the combustion of 0.25 mol of $C_2H_5OH(g)$?

(A) 5 L (B) 8 L (C) 10 L (D) 15 L

18. Adipic acid, $HOOC(CH_2)_4COOH$, is used in making nylon. What is the total number of atoms in 1.0 g of adipic acid?

Molar Mass, $g.mol^{-1}$	
adipic acid	146.26

(A) 20 (B) 4.1×10^{21} (C) 8.2×10^{22} (D) 7.2×10^{24}

19. Calculate the mass percentage of nitrogen in hydrazinium sulfate $(N_2H_5)_2SO_4$.

Molar Mass, $g.mol^{-1}$	
$(N_2H_5)_2SO_4$	162.2

(A) 10.8 (B) 17.3 (C) 34.5 (D) 51.2

20. Acetylene, C_2H_2, reacts with oxygen according to the unbalanced equation:

$C_2H_2\ (g) + O_2\ (g) \rightarrow CO_2\ (g) + H_2O(g)$

What is the O_2/C_2H_2 ratio when this equation is correctly balanced?

(A) 2/1 (B) 3/1 (C) 4/1 (D) 5/2

21. Silicon carbide, SiC, is produced by heating SiO_2 and C to high temperatures according to the equation:

$SiO_2(s) + 3C(s) \rightarrow SiC(s) + 2CO(g)$

How many grams of SiC could be formed by reacting 2.00 g of SiO_2 and 2.00 g of C?

(A) 1.33 (B) 2.26 (C) 3.59 (D) 4.00

22. A 7.66 g sample of hydrated sodium sulfate, $Na_2SO_4.xH_2O$, forms 4.06 g of anhydrous Na_2SO_4. What is the value of x?

Molar Mass, $g.mol^{-1}$	
Na_2SO_4	142

(A) 0.2 (B) 3.6 (C) 5 (D) 7

23. Silver metal reacts with nitric acid according to the equation:

$3Ag(s) + 4HNO_3(aq) \rightarrow 3AgNO_3(aq) + NO(g) + 2H_2O(l)$

What volume of 1.15 M $HNO_{3(aq)}$ is required to react with 0.784 g of silver?

(A) 4.74 mL (B) 6.32 mL (C) 8.43 mL (D) 25.3 mL

24. How many water molecules are in a 0.10 g sample of $CuSO_4.5H_2O$ (MM = 249.7)?

(A) 1.2×10^{21} (B) 2.4×10^{21} (C) 2.4×10^{22} (D) 1.2×10^{23}

25. Acetylene, C_2H_2, reacts with O_2 to produce CO_2 and H_2O. What is the O_2/C_2H_2 ratio in the balanced equation?

(A) 2/1 (B) 3/2 (C) 5/2 (D) 3/1

26. How many moles of stomach acid can be neutralized by $Mg(OH)_2$ in the form of Milk of Magnesia is used to neutralize excess stomach acid 1.00 g of $Mg(OH)_2$?

(A) 0.0171 (B) 0.0343 (C) 0.686 (D) 1.25

27. A 25.00 mL sample of 0.1050 M H_2SO_4 is titrated with a NaOH solution of unknown concentration. The phenolphthalein endpoint was reached when 17.23 mL of the NaOH solution had been added. What is the concentration of the NaOH?

(A) 0.07617 M (B) 0.1447 M (C) 0.1524 M (D) 0.3047 M

28. A sample of oxygen gas and a sample of an unknown gas are weighed separately in the same evacuated flask. Use the data given to find the molar mass of the unknown gas (assume experiments are carried out at the same pressure and temperature).

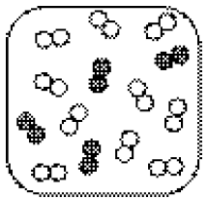
Mass of evacuated flask	124.46 g
Mass of flask + oxygen	125.10 g
Mass of flask + unknown gas	125.34 g

(A) 22 g/mol (B) 38 g/mol (C) 44 g/mol (D) 84 g/mol

29. What is the mass of one molecule of water in grams?

(A) 18 (B) 1.1×10^{-21} (C) 3.0×10^{-23} (D) 1.7×10^{-24}

30. If nitrogen atoms are represented as filled circles and oxygen atoms as open circles, how much NO_2 can be prepared from the mixture shown?



(A) 4 molecules (B) 5 molecules (C) 6 molecules (D) 8 molecules

31. Naturally occurring thallium consists of two stable isotopes, Tl-203 and Tl-205 (atomic masses = 203.0 and 205.0, respectively) and has an average atomic mass of 204.4. What is the percentage of Tl-205?

(A) 14.0% (B) 30.0% (C) 50.0 % (D) 70.0 %

32. What is the maximum mass (in grams) of NO that could be obtained from15.5 g of N_2O_4 and 4.68 g of N_2H_4 when they react? The balanced chemical equation is;

$2N_2O_4 + N_2H_4 \rightarrow 6NO + 2H_2O$.

(A) 4.38 (B) 5.04 (C) 15.2 (D) 26.2

33. What volume of 0.108 M H_2SO_4 is required to neutralize 25.0 mL of 0.145 M KOH?

(A) 16.8 mL (B) 33.6 mL (C) 37.2 mL (D) 67.1 mL

34. A mineral containing iron(II) sulfide but no other sulfides is treated with excess hydrochloric acid to produce hydrogen sulfide. If a 3.15 g sample of the mineral yields 448 mL of hydrogen sulfide gas (measured at 0 °C and 760 mm Hg), what is the mass percentage of iron(II) sulfide in the sample?

(A) 20.4 (B) 35.5 (C) 55.8 (D) 71.0

35. Iodine adds to the double bonds in fatty acids (one iodine molecule per double bond). How many double bonds are in a molecule of arachidonic acid (Molar mass = 304.5 g/mol) if 0.125 g of the acid require 0.417 g of iodine?

(A) 2 (B) 3 (C) 4 (D) 8

Question number 36, 37, 38 and 39 are based on the following information:

Dissolved oxygen in water is determined by using a redox reaction. Following equations describe the procedure -

I $2Mn^{2+}(aq) + 4OH^-(aq) + O_2(g) \longrightarrow 2MnO_2(s) + 2H_2O(\ell)$

II $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \longrightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(\ell)$

III $2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow 2S_4O_6^{2-}(aq) + 2I^-(aq)$

36. How many moles of $2S_2O_3^{2-}$ are equivalent to each mole of O_2?

(A) 0.5 (B) 1 (C) 2 (D) 4

37. What amount of I_2 will be liberated from 8 g dissolved oxygen?

(A) 127 g (B) 254 g (C) 504 g (D) 1008 g

38. 3×10^{-3} moles O_2 is dissolved per litre of water, then what will be molarity of I^- produced in the given reaction?

(A) 3×10^{-3} M (B) $4 \times 3 \times 10^{-3}$ M (C) $2 \times 3 \times 10^{-3}$ M (D) $\frac{1}{2} \times 3 \times 10^{-3}$ M

39. 8 mg dissolved oxygen will consume -

(A) 5×10^{-4} mol Mn^{+2} (B) 2.5×10^{-4} mol Mn^{2+}

(C) 10 mol Mn^{2+} (D) 2 mol Mn^{2+}

40. The equation: $2Al(S) + \frac{3}{2} O_2(g) \longrightarrow Al_2O_3(S)$ show that

(A) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce $\frac{7}{2}$ mole of Al_2O_3

(B) 2g of Al reacts with $\frac{3}{2}$ g of O_2 to produce one mole of Al_2O_3

(C) 2g mole of Al reacts with $\frac{3}{2}$ litre of O_2 to produce 1 mole of Al_2O_3

(D) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce 1 mole of Al_2O_3

41. Volume of CO_2 obtained at STP by the complete decomposition of 9.85 g. $BaCO_3$ is –
(At. wt. of Ba = 137)

(A) 2.24 lit (B) 1.12 lit (C) 0.84 lit (D) 0.56 lit

42. The volume of a gas in discharge tube is 1.12×10^{-7} ml. at STP then the number of molecule of gas in the tube is:

(A) 3.01×10^4 (B) 3.01×10^{15} (C) 3.01×10^{12} (D) 3.01×10^{16}

43. A person adds 1.71 gram of sugar ($C_{12}H_{22}O_{11}$) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)

(A) 3.6×10^{22} (B) 7.2×10^{21} (C) 0.05 (D) 6.6×10^{22}

44. The volume of CO_2 (in litres) liberated at STP when 10 g of 90% pure limestone is heated completely, is-

(A) 22.4 L (B) 2.24 L (C) 20.16 L (D) 2.016 L

45. The number of atoms of Cr and O in a compound are 4.8×10^{10} & 9.6×10^{10} respectively. Its empirical formula is-

(A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) None

46. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of that compound (its mol wt. is 120) could be

(A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2

47. 1 litre of a hydrocarbon weights as much as one litre of CO_2.Then the molecular formula of the hydrocarbon is:

(A) C_3H_8 (B) C_2H_6 (C) C_2H_4 (D) C_3H_6

48. In the reaction, $CrO_5 + H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + H_2O + O_2$
one mole of CrO_5 will liberate how many moles of O_2?
(A) 5/2 (B) 5/4 (C) 9/2 (D) None

49. The moles of O_2 required for reacting with 6.8 g of ammonia
(.....NH_3 + ... O_2 ⟶ ..... NO + ... H_2O) is
(A) 5 (B) 2.5 (C) 1 (D) 0.5

50. The hydrated salt, $Na_2SO_4.nH_2O$ undergoes 55.9% loss in weight on heating and becomes anhydrous. The value of n will be -
(A) 5 (B) 3 (C) 7 (D) 10

51. How many gram of KCl would have to be dissolved in 60 g of H_2O to give 20% by weight of solution -
(A) 15 g (B) 1.5 g (C) 11.5 g (D) 31.5 g

52. If V ml of the vapours of substance at NTP weight W g. Then mol w. of substance is:
(A) (W/V) × 22400 (B) $\frac{V}{W} \times 22.4$ (C) (W–V) × 22400 (D) $\frac{W \times 1}{V \times 22400}$

53. The molality of a sulphuric acid solution is 0.2. Calculate the total weight of the solution having 1000 gm of solvent.
(A) 1000 g (B) 1098.6 g (C) 980.4 g (D) 1019.6g

54. The total number of AlF_3 molecule in a sample of AlF_3 containing 3.01×10^{23} ions of F^- is -
(A) 9.0×10^{24} (B) 3.0×10^{24} (C) 7.5×10^{23} (D) 10^{23}

55. Equal masses of O_2, H_2 and CH_4 are taken in a container. The respective mole ratio of these gases in container is
(A) 1: 16: 2 (B) 16: 1: 2 (C) 1: 2: 16 (D) 16: 2: 1

56. The total number of molecules of SO_2Cl_2 in 13.5 g of sulphuryl chloride is
(A) 0.1 (B) 0.2 (C) 0.3 (D) 0.4

57. The number of sodium atoms in 2 moles of sodium ferrocyanide ($Na_4[Fe(CN)_6]$) is:
(A) 12 × 23 (B) 26×10^{23} (C) 34×10^{23} (D) 48×10^{23}

58. The empirical formula of a compound of molecular mass 120 is CH_2O. The molecular formula of the compound is
(A) $C_2H_4O_2$ (B) $C_4H_8O_1$ (C) $C_3H_6O_3$ (D) all of these

59. A hydrocarbon contains 75% of carbon. Then its empirical formula is:
(A) CH_4 (B) C_2H_4 (C) C_2H_6 (D) C_2H_2

60. Which of the following compounds has same empirical formula as that of glucose:
(A) CH_3CHO (B) CH_3COOH (C) CH_3OH (D) C_2H_6

61. Caffeine has a molecular weight of 194. It contains 28.9% by mass of nitrogen number of atoms of nitrogen in one molecule of it:

(A) 2 (B) 3 (C) 4 (D) 5

62. 10g of piece of marble was put into excess of dilute HCl acid. When reaction was complete 1120 cm^3 of CO_2 was obtained at STP. The evalute the value of x /10 where x = % of $CaCO_3$ in marble.

(A) 4 (B) 5 (C) 8 (D) 10

63. Vapour density of a gas if its density is 0.178 g/L at NTP is:

(A) 0.178 (B) 2 (C) 4 (D) 0.089

64. Vapour density of a volatile substance w.r.t. CH_4 is 4(CH_4 = 1). Its molecular weight would be –

(A) 8 (B) 32 (C) 64 (D) 128

65. A gas is found to have the formula (CO)x. It's VD is 70 the value of x must be:

(A) 7 (B) 4 (C) 5 (D) 6

66. 8g of sulphur are burnt to form SO_2, which is oxidised by Cl_2 water. The solution is treated with $BaCl_2$ solution. The amount of $BaSO_4$ precipitated is -

(A) 1.0 mole (B) 0.5 mole (C) 0.75 mole (D) 0.25 mole

67. Weight of oxygen in Fe_2O_3 and FeO is in the simple ratio of -

(A) 3: 2 (B) 1: 2 (C) 2: 1 (D) 3: 1

68. Amount of oxygen required for combustion of 1 kg of a mixture of butane and isobutane is -

(A) 1.8 kg (B) 2.7 kg (C) 4.5 kg (D) 3.58 kg

69. Amount of $BaSO_4$ formed on mixing the aqueous solution of 2.08 g $BaCl_2$ and excess of dilute H_2SO_4 is–

(A) 2.33 g (B) 2.08 g (C) 1.04 g (D) 1.165 g

70. The volume of oxygen required for complete combustion of 20 ml of ethene is

(A) 30 ml (B) 60 ml (C) 40 ml (D) 50 ml

71. The volume of gas at NTP produced by 100g of CaC_2 with water:

(A) 70 litre (B) 35 litre (C) 17.5 litre (D) 22.4 litre

72. Sulphur trioxide is prepared by the following two reactions

$S_8(s) + 8O_2(g) \rightarrow 8SO_2(g)$

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

How many grams of SO_3 are produced from 1 mol of S_8?

(A) 1280.0 (B) 640.0 (C) 960.0 (D) 320

73. Calculate the amount of H_2 which is left unreacted in the given reaction:

$2H_2 + O_2 \longrightarrow 2H_2O$

If 8 g of H_2 is mixed with 16 g O_2?

(A) 3 g (B) 6 g (C) 1 g (D) 4 g

74. A mixture of 1.0 mole of Al and 3.0 mole of Cl_2 are allowed to react as:

$2Al\ (s) + 3Cl_2\ (g) \longrightarrow 2AlCl_3\ (s)$

(a) Which is limiting reagent?

(b) How many moles of $AlCl_3$ are formed

(c) Moles of excess reagent left unreacted is

(A) (a) Al, (b) 1.0 (c) 1.5 (B) (a) Cl_2, (b) 2.0 (c) 2.0

(C) (a) Al, (b) 0.5 (c) 1.5 (D) (a) Cl_2, (b) 1.0 (d) 1.5

75. Molarity of H_2SO_4 (density 1.8g/mL) is 18M. The molality of this solution is -

(A) 36 (B) 200 (C) 500 (D) 18

76. The mole fraction of water in 20% (wt. /wt.) aqueous solution of H_2O_2 is -

(A) $\frac{77}{68}$ (B) $\frac{68}{77}$ (C) $\frac{20}{80}$ (D) $\frac{80}{20}$

77. When $FeCl_3$ is ignited in an atmosphere of pure oxygen, the following reaction takes plac

$4FeCl_3(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s) + 6Cl_2(g)$

If 3 moles of $FeCl_3$ are ignited in the presence of 2 moles of O_2 gas, how much of which reagent is present in excess and therefore, remains unreacted ?

(A) 0.33 mole $FeCl_3$ remains unreacted (B) 0.67 mole $FeCl_3$ remains unreacted

(C) 0.25 mole O_2 remains unreacted (D) 0.50 mole O_2 remains unreacted

78. The molarity of 98% H_2SO_4 (d = 1.8g/mL) by wt. is -

(A) 6 M (B) 18.74 M (C) 10 M (D) 4 M

79. If 500 ml of 1 M solution of glucose is mixed with 500 ml of 1 M solution of glucose final molarity of solution will be:

(A) 1 M (B) 0.5 M (C) 2 M (D) 1.5 M

80. What volume of a 0.8 M solution contains 100 millimoles of the solute?

(A) 100 mL (B) 125 mL (C) 500 mL (D) 62.5 mL

81. 500 mL of a glucose solution contains 6.02×10^{22} molecules. The concentration of the solution is

(A) 0.1 M (B) 1.0 M (C) 0.2 M (D) 2.0 M

82. What is the concentration of nitrate ions if equal volumes of 0.1 M $AgNO_3$ and 0.1 M NaCl are mixed together:

(A) 0.1 M (B) 0.2 M (C) 0.05 M (D) 0.25 M

83. The molarity of the solution containing 2.8% mass-volume solution of KOH is

(A) M/10 (B) M/2 (C) M/5 (D) 1 M

84. The mole fraction of water in a solution containing 117 g sodium chloride and 900 g of water is?

(A) 0.0632 (B) 0.038 (C) 0.9615 (D) 1.000

85. 250 ml of 0.5 M KCl is diluted with water to 500 ml of solution, the number of chloride ions in the resulting solution are

(A) 6.02×10^{23} (B) 3.76×10^{22} (C) 1×10^{24} (D) 3.76×10^{23}

NSEJS PYQ

1. A chemist's report on a batch of pharmaceutical products, Aspirin ($C_9H_8O_4$) (250mg tablets) and Paracetamol ($C_8H_9NO_2$) (500mg tablets) indicated a '+0.5%' weight error in each tablet. Due to this error, the consumer gets extra 'x' molecules of aspirin per tablet and extra 'y' molecules of paracetamol per tablet. Choose the 'best' relation between x and y. **[NSEJS 2015]**

(A) x = y (B) x > y (C) y > x (D) x = 2y

2. Sodium reacts with excess oxygen to form sodium oxide. A student wants to prepare 1.24g of sodium oxide. While doing the calculations, he uses atomic number of sodium instead of atomic mass. What is the approximate percentage error in the mass of sodium oxide obtained due to this mistake?**[NSEJS 2015]**

(A) 11% (B) 23% (C) 48% (D) 60%

3. Sati was studying neutralisation reaction. She accidentally dropped 'x' grams of a sodium carbonate monohydrate into a 100 mL solution of HCl whose concentration was 0.25 M. 10mL from this final solution was taken and titrated against 0.05M NaOH solution to yield a titre value of 20mL. What is the value of 'x'. **[NSEJS 2015]**

(A) 186mg (B) 93mg (C) 1860mg (D) 930mg

4. What is the ratio of number of electrons gained by acidified $KMnO_4$ and acidified $K_2Cr_2O_7$ in the reaction given below **[NSEJS 2015]**

$KMnO_4 \xrightarrow{H^+} Mn^{2+}$

$K_2Cr_2O_7 \xrightarrow{H^+} Cr^{3r}$

(A) 5 : 6 (B) 6 : 5 (C) 3 : 5 (D) 5 : 3

5. 5 g of a gas at 27°C occupied the same volume as 0.250 g of dihydrogen at 17°C at the same pressure. Find out molecular weight of the gas. **[NSEJS 2015]**

(A) 41.37 (B) 20.68 (C) 25.80 (D) 39.87

6. Air has three major components: nitrogen, oxygen, and argon. Given that one mole of air at sea level is made up of 78% nitrogen, 21% oxygen, and 1% argon, by volume. What is the density of air? Assume that gases behave in an ideal manner. (Atomic mass of argon is 40 amu) **[NSEJS 2016]**

(A) 14.62 g/L (B) 1.3 g/L (C) 29 g/L (D) 0.65 g/L

7. If 0.50 mole of a monovalent metal (M^{+1}) halide is mixed with 0.2 mole of a divalent metal (L^{+2}) phosphate, the maximum number of moles of M_3PO_4 that can be formed is **[NSEJS 2016]**

(A) 0.25 (B) 0.30 (C) 0.16 (D) 0.20

8. Every major city in India has a pollution control board to monitor air and water pollution. The following data is from three different localities in Bangalore city from the year 2015. **[NSEJS 2016]**

Locality	Annual average of SO_2 in the air (volume/volume)
P	16.3 mL/m^3
Y	16.3 ppb (m^3/m^3)
Z	16.3 ppm (m^3/m^3)

ppb stands for parts per billion and ppm stands for parts per million. These are different units to express concentration. They are very similar to percentage (which is actually parts per hundred.)

Based on the above data, which place will you choose to live in?

(A) All localities are equally polluted, so I have no preference.

(B) P is the more polluted than Y and Z, hence I will live in either Y or Z.

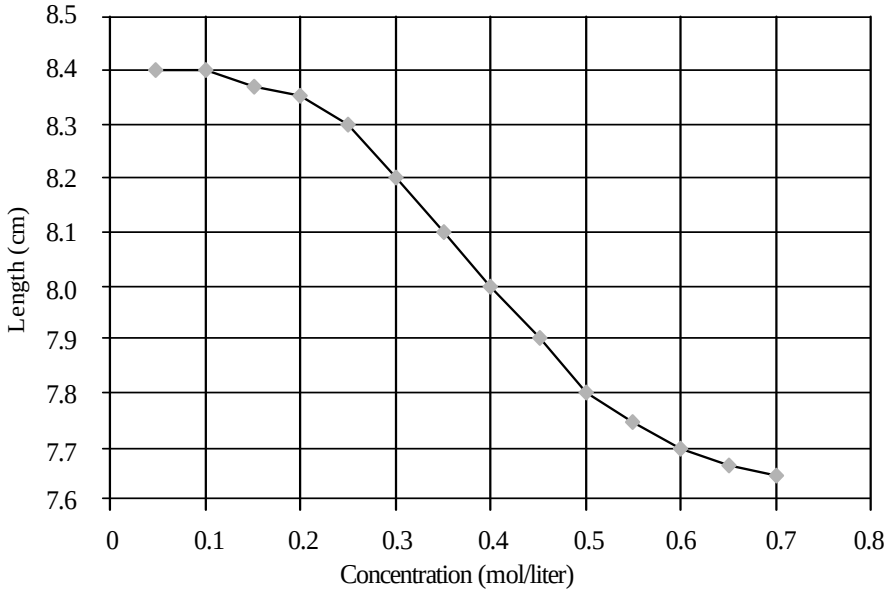
(C) Locality Y is least polluted, hence I will live in Y.

(D) Z and Y are more polluted than P, hence I will live in P.

9. When 1 gram of a mixture of aluminium and zinc was treated with HCl, a gas was liberated. At the end of the reaction, the volume of the liberated gas was found to be 524 cm^3, under STP conditions. The individual weights of aluminium and zinc in the mixture, respectively, are: **[NSEJS 2016]**

(A) 0.2g and 0.8g (B) 0.8g and 0.2g (C) 0.5g and 0.5g (D) 0.322g and 0.678g

10. Thin cuboidal strips are made by slicing a potato. They are all made to be exactly 8 cm long and 2 mm wide. Each strip is placed in sugar solutions of different concentration. After soaking it for 5 hours, their lengths are measured again. The following graph shows the results of the experiment. What concentration of sugar solution is isotonic with the contents of the cells of the potato. **[NSEJS 2016]**



(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.1

11. When solid KOH is mixed with solid NH_4Cl, a gas is produced. Which gas is it? **[NSEJS 2016]**

(A) Chlorine (B) hydrogen (C) hydrogen chloride (D) ammonia

12. A substance is dissolved in water, forming a 0.5 molar solution. If 4.0 L of solution contains 240g of the substance, what is the molecular mass of the substance? **[NSEJS 2018]**

(A) 60g/mole (B) 120g/mole (C) 240g/mole (D) 480g/mole

13. A car battery was kept for charging and after getting fully charged density of the battery acid (H_2SO_4) was measured and found to be1.28 g cm^{-3}. If Initial molarity of battery acid was 4.2 M then mass percentage will be around_____. **[NSEJS 2018]**

(A) 28% (B) 30% (C) 32% (D) 34%

14. Element "X" with atomic mass 10 was allowed to react completely with element "Y" of atomic mass 20 to form a compound. When this compound was analysed it was found that it contains 60% of X and 40 % of Y by weight. The simplest formula of this compound will be_____. **[NSEJS 2018]**

(A) X_3Y (B) X_2Y_3 (C) Y_3X (D) X_6Y_4

15. 4.095 X 10^{24} nitrogen atoms are filled in an enclosed gas cylinder of capacity two litre. The number of moles of nitrogen gas in the cylinder is___. **[NSEJS 2018]**

(A) 14.7 (B) 6.8 (C) 3.4 (D) 2.9

16. A solution of pure aluminium sulphate containing 0.170g of aluminium ions is treated with excess of barium hydroxide solution. Total weight of the precipitate will be: **[NSEJS 2018]**

(A) 0.5g (B) 2.7g (C) 1.7 g (D) 0.54g

17. W g of pure coal was combusted in pure dry oxygen. The carbon dioxide gas obtained was absorbed in 0.1M KOH solution. The complete absorption of CO_2 required 5 cm^3 of 0.1M KOH. The amount of coal combusted is **[NSEJS 2018]**

(A) 3mg (B) 6mg (C) 11mg (D) 12mg

18. Sulphur di-oxide gas and ammonia gas were mixed in different proportions. The pair of gases containing same number of molecules at NTP is _____. **[NSEJS 2018]**

(A) 1120 cm^3 of SO_2 + 0.85 g of ammonia (B) 0.25g mole of SO_2 + 2240 cm^3 of ammonia

(C) 1680 cm^3 of SO_2 + 1.7 g of ammonia (D) 0.25 g mole of SO_2 + 0.85 g of ammonia

19. A strip of iron with mass 15.5 g is placed in a solution containing 21.0 g copper sulphate. After some time the reaction stops. Iron strip was found to have mass 8.5 g. The mass of copper formed was found to be 8.60g. Find the mass of ferrous sulphate formed in this reaction. **[NSEJS 2018]**

(A) 19.40 g (B) 18.40 g (C) 17.40 g (D) 16.40 g

20. Sonu has N/2 HCl solution and Monu has N/10 HCl solution. They are asked to prepare 2 litres of N/5 HCl solution. What volume of two solutions be mixed? **[NSEJS 2018]**

(A) (0.5 +1.5) litre (B) (1.0 +1.0) litre (C) (0.3 +1.7) litre (D) (0.2 +1.8) litre

21. A solution (P) was prepared by dissolving 6.3 g of oxalic acid in 100 ml water. 25 ml of this solution was taken and was further diluted to 250 ml to prepare solution (Q). What weight of NaOH in ppm will be required to neutralize 10 ml of solution (Q)? **[NSEJS 2018]**

(A) 10 ppm (B) 20 ppm (C) 40 ppm (D) 80 ppm

22. $2KBrO_3 + 12H^+ + 10e^- \rightarrow Br_2 + 6H_2O + 2K^+$

From above reaction the equivalent weight of $KBrO_3$ can be calculated as (M is molecular weight of $KBrO_3$) **[NSEJS 2018]**

(A) M/5 (B) M/10 (C) M/12 (D) M/2

23. Four gram of mixture of calcium carbonate and sand is treated with excess of HCl and 0.880 g of carbon-di-oxide is produced. What is the percentage of calcium carbonate in original mixture? **[NSEJS 2019]**

(A) 40 % (B) 50% (C) 55% (D) 45%

24. Substance X is white crystalline solid which melts after 10 seconds on burner flame. It is soluble in water and insoluble in CCl_4. It is a poor conductor of electricity in molten state as well as in the form of aqueous solution, hence we conclude that substance X is **[NSEJS 2019]**

(A) an ionic compound (B) a non polar covalent compound

(C) a polar covalent compound (D) a pure element

25. In a beaker 50 ml of a normal HCl solution was taken and NH_3 gas was passed through it for some time. The contents of the beaker were then titrated, which required 60 ml of semi normal NaOH solution. How much ammonia was passed through the beaker? **[NSEJS 2019]**

(A) 0.85 g (B) 0.34 g (C) 0.51 g (D) 0.4 g

26. What is the ratio of reducing agent to oxidizing agent, if the following reaction is correctly balanced? **[NSEJS 2019]**

$NH_3 + O_2 \rightarrow NO + H_2O$

(A) 4:5 (B) 5:4 (C) 5:3 (D) 3:5

27. In one litre of pure water, 44.4 g of calcium chloride is dissolved. The number of ions in one mL of the resultant solution is: **[NSEJS 2019]**

(A) 7.23×10^{23} (B) 7.23×10^{20} (C) 4.82×10^{23} (D) 4.82×10^{20}

28. A zinc rod was dipped in 100 cm^3 of 1M copper chloride solution. After certain time the molarity of Cu^{2+} ions in the solution was found to be 0.8 M. If the weight of zinc rod is 20 g, then the molarity of chloride ions is_____. **[NSEJS 2019]**

(A) 2 M (B) 1.5 M (C) 1 M (D) 0.5 M

29. Which of the following species is / are isoelectronic with Neon? **[NSEJS 2019]**

(i) N^{3-} (ii) Mg^{2+} (iii) K^+ (iv) Ca^{2+}

(A) only (iv) (B) only (ii) (C) both (i) & (ii) (D) both (i) & (iii)

30. Which of the following gases will have equal volume at STP, if the weight of gases is 14.0 g? **[NSEJS 2019]**

(i) N_2O (ii) NO_2 (iii) N_2 (iv) CO

(A) (i) & (ii) (B) (ii) & (iii) (C) (i) & (iii) (D) (iii) & (iv)

31. A nuclear reactor is working at 30% efficiency (i.e. conversion of nuclear energy to electrical energy). In this reactor $^{235}_{92}U$ nucleus undergoes fission and releases 200 MeV energy per atom. If 1000 kW of electrical power is obtained in this reactor, then the number of atoms disintegrated (undergone fission) per second in the reactor is **[NSEJS 2020]**

(A) 1.04×10^{17} (B) 6.5×10^{12} (C) 3.125×10^{12} (D) 3.25×10^{32}

32. In one process for waterproofing, a fabric is exposed to $(CH_3)_2SiCl_2$ vapors. The vapors react with the hydroxyl groups on the surface of the fabric or with traces of water to form the waterproofing film $[(CH_3)_2SiO]_n$, by the reaction; $n(CH_3)_2SiCl_2 + 2nOH^- \rightarrow 2nCl^- + nH_2O + \left[(CH_3)_2SiO\right]_n$ where n stands for a larger integer. The waterproofing film is deposited on the fabric layer upon layer. Each layer is 6 Å thick [the thickness of the $(CH_3)_2SiO$ group]. How much $(CH_3)_2SiCl_2$ is needed to waterproof one side of a piece of fabric, 1m by 2m, with a film 300 layers thick? The density of the film is 1.0 g/cm^3. **[NSEJS 2020]**

(A) 0.63g (B) 0.36g (C) 6.3g (D) 3.6g

33. Triclosan ($C_{12}H_7Cl_3O_2$) is an antibacterial and antifungal agent. It is a polychloro phenoxy phenol. It is widely used as a preservative and antimicrobial agent in personal care products such as soaps, skin creams, and deodorants etc. A label on a 200 mL hand sanitizer bottle claims that it contain Triclosan 0.2% w/v. What will be the number of molecules of Triclosan present in the bottle? **[NSEJS 2020]**
(N_A is Avogadro's Number)

(A) 1.4 X 10^{25} N_A (B) 1.4 X 10^{24} N_A

(C) 1.4 X 10^{23} N_A (D) 1.4 X 10^{22} N_A

34. Suppose that A and B forms compound B_2A_3 and B_2A. If 0.05 mole of B_2A_3 weighs 12 g and 0.1 mole of B_2A weighs 10 g, what are the atomic weight of A and B respectively? **[NSEJS 2020]**

(A) 70 and 25 (B) 50 and 20 (C) 40 and 30 (D) 30 and 40

35. The ratio of the charge of an ion or subatomic particle to its mass (q/m) is called specific charge. Then the correct options are **[NSEJS 2020]**

(A) SI unit of specific charge can be written as A·s /kg.
(B) If all the isotopes of hydrogen are ionized then tritium will have least specific charge among them.
(C) specific charge of an α- particle will be greater than that of an electron.
(D) specific charge ratio of an electron is 1.75×10^{11} C/kg.

36. Coal is a common fossil fuel. It contains 0.2 to 5.0 percent sulphur which on burning produces a gas responsible for acid rain. The number of atoms in one mole of this gas is- **[NSEJS 2021]**

(A) 6.02×10^{23} (B) 1.81×10^{23} (C) 1.81×10^{24} (D) 1.21×10^{24}

37. The stomach fluid in human contains HCl, KCl and NaCl. The stomach fluid is highly acidic and plays an important role in the digestion of food as well as killing of bacteria. The increased acidity may lead to abdominal pain, nausea, bloating, and heartburn. Such a patient is prescribed antacid tablet which mainly contains aluminium hydroxide (Mol. Wt. 78). If the concentrations of HCl, KCl and NaCl are 0.01M each and the stomach fluid volume is 2 litre, the amount of $Al(OH)_3$ required to neutralize the fluid will be- **[NSEJS 2021]**

(A) 0.52 g (B) 1.08 g (C) 0.81 g (D) 2.16 g

38. A 0.500 g mixture of calcium carbonate and calcium oxide was strongly heated to produce a non-combustible gas. If the weight of the residue obtained on heating is found to be 0.434 g, the percentage of calcium oxide in the mixture is- **[NSEJS 2021]**

(A) 70% (B) 30% (C) 35% (D) 60%

39. Equal masses of two gases among N_2, NO, O_2, CO, CO_2 and SO_2 occupy same volume at STP. These two gases are: **[NSEJS 2021]**

(A) N_2 and O_2 (B) CO and NO (C) SO_2 and CO_2 (D) N_2 and CO

40. Soaps and detergents are common agents used in laundry industry. They are long chain hydrocarbons with ionic terminals of cationic or anionic nature. A 1% (w/v) soap solution X and 1% (w/v) detergent solution Y were prepared in distilled water. Each of the solutions was divided in two equal parts and labelled as X_1, X_2, Y_1 and Y_2. 1 g NaCl was added to X_1 and Y_1 each while 1 g $CaCl_2$ was added to X_2 and Y_2 each. Which of the following observations is/are correct? **[NSEJS 2021]**

(A) X_1 shows slimy precipitate (B) X_2 shows slimy precipitate
(C) Y_1 shows slimy precipitate (D) Y_2 shows slimy precipitate

41. 1 kg of aqueous urea solution (mole fraction of solute = 0.2) is diluted to 5 kg. Mole fraction of solute in diluted solution is: **[NSEJS 2022]**

(A) 0.2 (B) 4×10^{-2} (C) 0.029 (D) 0.971

42. What is the percentage of $MgCO_3$ in mixture of $MgCO_3$ and $CaCO_3$ if its 2g H_2SO_4 for complete neutralization? **[NSEJS 2022]**

(A) 89% (B) 11% (C) 50% (D) 25%

43. A mixture of HCOOH and $H_2C_2O_4$ is heated with Conc. H_2SO_4. The gases produced were passed through KOH solution where their volume decreased by $\frac{1}{6}$. Ration of two acids in the mixture: **[NSEJS 2022]**

(A) 1 : 4 (B) 4 : 1 (C) 1 : 1 (D) data insufficient

44. Which state of matter exists at very high temperature and at very low temperature (near absolute zero) respectively? BEC stands for Bose Einstein Condensate. **[NSEJS 2023]**

(A) BEC, fermionic condensate (B) Plasma, BEC

(C) Fermionic condensate, Plasma (D) Gas, BEC

45. An element X has two natural isotopes: $^{10}_{5}X$ (atomic mass 10.013 u) and $^{11}_{5}X$ (atomic mass 11.009 u). Relative abundance of these isotopes in nature has been recorded 19.8 % and 80.2 % respectively. On the basis of these data, average atomic mass of element X is close to **[NSEJS 2023]**

(A) 10.210 u (B) 10.511 u (C) 10.799 u (D) 10.812 u

46. A boy gifted a diamond ring to his mother on her wedding anniversary. If this diamond ring contains 3 carat diamond then number of carbon atoms he gifted to his mother is **[NSEJS 2023]**

Given – (1 carat = 200 mg)

(A) 3.01×10^{23} (B) 2.1×10^{23} (C) 3.01×10^{22} (D) 2.1×10^{22}

47. A sample of 1.0 liter milk weighs 1.032 kg. The milk contains 4% milk fat by volume having density 865 kg/m^3. The density of fat free (skimmed) milk in kg/m^3 is **[NSEJS 2024]**

(A) 1036.95 (B) 1037.95 (C) 1038.95 (D) 1040.95

48. A research scholar requires 50 milliliter aqueous $NaNO_3$ solution containing 70.0 mg of Na^+ per milliliter. The amount of $NaNO_3$ required for this purpose is **[NSEJS 2024]**

(A) 0.350 g (B) 0.161 g (C) 29.75 g (D) 12.94 g

49. How many moles of the so called 'electron gas' weighs one kilogram? **[NSEJS 2024]**

(A) $\frac{6.02}{9.11}\times10^{54}$ (B) $\frac{100}{6.02\times9.11}\times10^{6}$ (C) 6.02×10^{23} (D) $\frac{10}{9.11}\times10^{30}$

50. The molecular formula of a commercial resin used for exchanging ions in water is $C_8H_7SO_3Na$. The maximum uptake of Ca^{2+} ions by the resin expressed in mol/g is **[NSEJS 2024]**

(A) $\frac{2}{309}$ (B) $\frac{1}{412}$ (C) $\frac{1}{103}$ (D) $\frac{1}{206}$

51. For Cl-atom, IE_1 and ΔH_{EG_1} values are 13.0 eV and 3.60 eV respectively. If Avogadro number of atoms are converted into Cl^- as $Cl + e^- \longrightarrow Cl^-$ and the energy released is completely used to ionize Cl-atom as $Cl \rightarrow Cl^+ + e^-$, then number of Cl-atom converted into Cl^+ are **[NSEJS 2024]**

(A) 1.667×10^{23} (B) 6.023×10^{23} (C) 3.2×10^{23} (D) can't be calculated

Advanced & Olympiad Challenger Questions

IJSO Stage-I

1. The volume of 0.5 M aqueous NaOH solution required to neutralize 10 ml of 2 M aqueous HCl solution is: **[KVPY-Part-I-2008]**

(A) 20ml (B) 40ml (C) 80ml (D) 120ml

2. 3.01×10^{23} molecules of elemental Sulphur will react with 0.5 mole of oxygen gas completely to produce **[KVPY-Part-I-2008]**

(A) 6.02×10^{23} molecules of SO_3 (B) 6.02×10^{23} molecules of SO_2

(C) 3.01×10^{23} molecules of SO_3 (D) 3.01×10^{23} molecules of SO_2

3. The density of a salt solution is1.13 g cm^{-3} and it contains 18% of NaCl by weight. The volume of the solution containing 36.0 g of the salt will be: **[KVPY-Part-II-2008]**

(A) 200 cm^3 (B) 217 cm^3 (C) 177 cm^3 (D) 157cm^3

4. One mole of nitrogen gas on reaction with 3.01×10^{23} molecules of hydrogen gas produces- **[KVPY-Part-I-2009]**

(A) one mole of ammonia (B) 2.0×10^{23} molecules of ammonia

(C) 2 moles of ammonia (D) 3.01×10^{23} molecules of ammonia

5. 10 ml of an aqueous solution containing 222 mg of calcium chloride (mol. wt. = 111) is diluted to 100 ml. The concentration of chloride ion in the resulting solution is - **[KVPY-Part-II-2009]**

(A) 0.02 mol/lit. (B) 0.01 mol/lit. (C) 0.04 mol/lit (D) 2.0 mol/lit.

6. Aluminium reduces manganese dioxide to manganese at high temperature. The amount of aluminium required to reduce one gram mole of manganese dioxide is - **[KVPY-Part-II-2009]**

(A) 1/2 gram mole (B) 1 gram mole (C) 3/4 gram mole (D) 4/3 gram mole

7. The molar mass of $CaCO_3$ is 100 g. The maximum amount of carbon dioxide that can be liberated on heating 25 g of $CaCO_3$ is - **[KVPY-Part-I-2010]**

(A) 11 g (B) 55 g (C) 22 g (D) 2.2 g

8. The weight of calcium oxide formed by burning 20 g of calcium in excess oxygen is: **[KVPY-Part-I-2012]**

(A) 36 g (B) 56 g (C) 28 g (D) 72 g

9. By dissolving 0.35 mole of sodium chloride in water, 1.30 L of salt solution is obtained. The molarity of the resulting solution should be reported: **[KVPY-Part-I-2012]**

(A) 0.3 (B) 0.269 (C) 0.27 (D) 0.2692

10. The weight percent of sucrose (formula weight = 342 g mol^{-1}) in an aqueous solution is 3.42. The density of the solution is 1 g mL^{-1}, the concentration of sucrose in the solution in mol L^{-1} is: **[KVPY-Part-I-2013]**

(A) 0.01 (B) 0.1 (C) 1.0 (D) 10

11. The amount of metallic Zn (Atomic weight = 65.4) required to react with aqueous sodium hydroxide to produce 1 g of H_2, is **[KVPY-Part-II-2013]**

(A) 32.7 g (B) 98.1 g (C) 65.4 g (D) 16.3 g

12. Natural abundances of $_{12}C$ and $_{13}C$ isotopes of carbon are 99% and 1%, respectively. Assuming they only contributes to the mol. wt. of C_2F_4, the percentage of C_2F_4 having a molecular mass of 101 is: **[KVPY-Part-II-2013]**

(A) 1.98 (B) 98 (C) 0.198 (D) 99

13. The volume of oxygen at STP required to burn 2.4 g of carbon completely is: **[KVPY-Part-I-2014]**

(A) 1.12 L (B) 8.96L (C) 2.24 L (D) 4.48L

14. 10 moles of a mixture of gydogen and oxygen gases at a pressure of 1 atm at constant volume and temperature react to form 3.6 g of liquid water. The pressure of the resulting mixture will be closest to **[KVPY-Part-II-2014]**

(A) 1.07 atm (B) 0.97 atm (C) 1.02 atm (D) 0.92 atm

15. The percentage of nitrogen by mass in ammonium sulphate is closest to
(atomic masses H = 1, N = 14, O = 16, S = 32) **[KVPY-Part-I-2015]**

(A) 21% (B) 24% (C) 36% (D) 16%

16. If Avogadro's number is A_0, the number of sulphur atoms present in 200 mL of 1N H_2SO_4 is: **[KVPY-Part-I-2015]**

(A) $A_0/5$ (B) $A_0/2$ (C) $A_0/10$ (D) A_0

17. When 262 g of xenon (atomic mass = 131) reacted completely with 152 g of fluorine (atomic mass = 19), a mixture of XeF_2 and XeF_6 was produced. The molar ratio XeF_2: XeF_6 is: **[KVPY-Part-II-2015]**

(A) 1: 2 (B) 1: 4 (C) 1: 1 (D) 1: 3

18. When 22.4 L of C_4H_8 at STP is burnt completely, 89.6 L of CO_2 gas at STP and 72 g of water are produced. The volume of the oxygen gas at STP consumed in the reaction is closest to: **[KVPY-Part-II-2015]**

(A) 89.6 L (B) 112 L (C) 134.4 L (D) 22.4 L

19. X g of Ca [atomic mass = 40] dissolves completely in concentrated HCl solution to produce 5.04 L of H_2 gas at STP. The value of X is closest to **[KVPY-Part-I-2017]**

(A) 4.5 (B) 8.1 (C) 9.0 (D) 16.2

20. In a closed vessel at STP, 50 L of CH_4 is ignited with 750 L of air (containing 20% O_2). The number of moles of O_2 remaining in the vessel on cooling to room temperature is closest to **[KVPY-Part-I-2017]**

(A) 5.8 (B) 2.2 (C) 4.5 (D) 6.7

21. The reaction of an alkene X with bromine produces a compound Y, which has 22.22% C, 3.71% H and 74.07% Br. The ozonolysis of alkene X gives only one product. The alkene X is:
[Given: atomic mass of C = 12; H = 1; Br = 80] **[KVPY-Part-II-2017]**
(A) ethylene (B) 1-butene (C) 2-butene (D) 3-hexene

22. The ratio of amounts of H_2S needed to precipitate all the metal ions from 100ml of 1 M $AgNO_3$ and 100 ml of 1m $CuSO_4$ will be **[IJSO 2008-2009]**
(A) 1: 2 (B) 2: 1 (C) 1: 1 (D) indefinite

23. 6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is: **[IJSO 2008-2009]**
(A) 0.001 M (B) 0.01 M (C) 0.02 M (D) 0.1 M

24. A 3.7 litre 1.0 M NaOH solution is mixed with another 5 litre 0.30 M NaOH solution. The molarity of the resulting solution is: **[IJSO 2009-2010]**
(A) 0.80 M (B) 1.10 M (C) 0.73 M (D) 0.59 M

25. One mole of oxalic acid is equivalent to: **[IJSO 2009-2010]**
(A) 0.5 mole of NaOH (B) 1 mole of NaOH (C) 1.5 mole of NaOH (D) 2 mole of NaOH

26. When 2.8 g of nitrogen reacts with 0.6 g of hydrogen, 3.4 g of ammonia gas is of produced. If 5.8 g of nitrogen is allowed to react with 0.6 g of hydrogen, then the amount of ammonia produced is: **[IJSO 2010-2011]**
(A) 1.7 g (B) 3.4 g (C) 4.7 g (D) 7.4 g

27. Number of molecule present in a given amount of CO_2 and O_2 is the same at STP. If the weight of CO_2 is 5.5 g, then the weight of O_2 is **[IJSO 2010-2011]**
(A) 5.5 g (B) 4.4 g (C) 4.0 g (D) 2.0 g

28. The mass of one molecule of phosphorus (^{31}P) is **[IJSO 2010-2011]**
(A) 31 g (B) 2.059×10^{-22} g (C) 5.147×10^{-23} g (D) 124 g

29. The compound used to remove carbon dioxide from air is **[IJSO 2011-2012]**
(A) Sodium carbonate (B) sodium hydroxide (C) sodium nitrate (D) sodium chloride

30. 8 Grams of oxygen at NTP contain **[IJSO 2012-2013]**
(A) 1.5×10^{23} molecules (B) 3.0×10^{23} molecules
(C) 6.023×10^{23} molecules (D) 1.5×10^{22} molecules

31. When 1g of $CaCO_3$ reacts with 50 ml of 0.1 M HCl, the volume of CO_2 produced is: **[IJSO 2012-2013]**
(A) 11.2 mL (B) 22.4 mL (C) 112 mL (D) 56 mL

32. How many H atoms are in 3.4 g of $C_{12}H_{22}O_{11}$? **[IJSO 2013-2014]**

(A) 6.0×10^{23} (B) 1.3×10^{23} (C) 3.8×10^{22} (D) 6.0×10^{23}

33. What is the molality of a solution made by dissolving 100 g of bromothymol blue ($C_{27}H_{28}Br_2O_5S$) in 1.00 L of ethanol on a winter's day at 10ºC? The density of ethanol at this temperature is 0.7979 kg L^{-1}: **[IJSO 2013-2014]**

(A) 0.100 mol kg^{-1} (B) 0.128 mol kg^{-1} (C) 0.160 mol kg^{-1} (D) 0.201 mol kg^{-1}

34. The mass of 0.2 mole of Oxygen molecule is: **[IJSO 2013-2014]**

(A) 6.4g (B) 3.2g (C) 1.6g (D) 7.75g

35. In the reaction, **[IJSO 2014-2015]**

$2KClO_3 \rightarrow 2KCl + 3O_2$

What is the volume of oxygen released under NTP conditions when 36.75g of $KClO_3$ is heated?

(A) 3.6 litres (B) 7.2 litres (C) 18 litres (D) 10 litres

36. Sulphuric acid is manufactured by the contact process in which sulphur dioxide reacts with oxygen in presence of a catalys. If 5.6 moles of SO_2 reacts with 4.8 moles of O_2 and a large excess of water, the maximum number of moles of H_2SO_4 that can be obtained is: **[IJSO 2014-2015]**

(A) 11.2 (B) 5.6 (C) 4.8 (D) 1.4

37. Which of the following contains the same number of atoms as 13.5 grams of aluminium? **[IJSO 2014-2015]**

(A) 20 g of calcium (B) 10g of magnesium

(C) 20 g of potassium (D) 10 of sodium

38. A chemist's report on a batch of pharmaceutical products, Aspirin ($C_9H_8O_4$) (250 mg tablets) and paracetamol ($C_8H_9NO_2$) (500 mg tablets) indicated a '+0.5% weigh error in each tablet. Due to this error, the consumer gets extra 'x' molecules of aspirin per tablet and extra 'y' molecules of paracetamol per tablet, Choose the 'best' relation between x and y **[IJSO 2015-2016]**

(A) x = y (B) x > y (C) y > x (D) x = 2y

39. Sodium reacts with excess oxygen to form sodium oxide. A student wants to prepare 1.24g of sodium oxide. While doing the calculations, he uses atomic number of sodium instead of atomic mass. What is the approximate percentage error in the mass of sodium oxide obtained due to this mistake? **[IJSO 2015-2016]**

(A) 11% (B) 23% (C) 48% (D) 60%

40. It 0.50 mole of a monovalent metal (M^{+1}) halide is mixed with 0.2 mole of a divalent metal (L^{+2}) phosphate, the maximum number of moles of M_3PO_4 that can be formed is: **[IJSO 2016-2017]**

(A) 0.25 (B) 0.30 (C) 0.16 (D) 0.20

41. When 1 gram of a mixture of aluminium and zinc was treated with HCl, a gas was liberated. At the end of the reaction, the volume of the liberated gas was found to be 524 cm^3, under STP conditions. The individual weights of aluminium and zinc in the mixture, respectively, are: **[IJSO 2016-2017]**

(A) 0.2g and 0.8g (B) 0.8g and 0.2g (C) 0.5g and 0.5g (D) 0.322g and 0.678g

42. A substance is dissolved in water, forming a 0.5 molar solution. If 4.0 L of solution contains 240g of the substance, what is the molecular mass of the substance? **[IJSO 2017-2018]**

(A) 60g/mole (B) 120g/mole (C) 240g/mole (D) 480g/mole

43. A car battery was kept for charging and after getting fully charged density of the battery acid (H_2SO_4) was measured and found to be 1.28 g cm^{-3}. If Initial molarity of battery acid was 4.2 M then mass percentage will be around _____. **[IJSO 2017-2018]**

(A) 28% (B) 30% (C) 32% (D) 34%

44. Element "X" with atomic mass 10 was allowed to react completely with element "Y" of atomic mass 20 to form a compound. When this compound was analysed it was found that it contains 60% of X and 40 % of Y by weight. The simplest formula of this compound will be_____. **[IJSO 2017-2018]**

(A) X_3Y (B) X_2Y_3 (C) Y_3X (D) X_6Y_4

45. A solution of pure aluminium sulphate containing 0.170g of aluminium ions is treated with excess of barium hydroxide solution. Total weight of the precipitate will be: **[IJSO 2017-2018]**

(A) 0.5g (B) 2.7g (C) 1.7 g (D) 0.54g

46. Sulphur di-oxide gas and ammonia gas were mixed in different proportions. The pair of gases containing same number of molecules at NTP is _____. **[IJSO 2017-2018]**

(A) 1120 $cm^3$of SO_2 + 0.85 g of ammonia (B) 0.25g mole of SO_2 + 2240 cm^3 of ammonia

(C) 1680 cm^3 of SO_2 + 1.7 g of ammonia (D) 0.25 g mole of SO_2 + 0.85 g of ammonia

47. A strip of iron with mass 15.5 g is placed in a solution containing 21.0 g copper sulphate. After some time the reaction stops. Iron strip was found to have mass 8.5 g. The mass of copper formed was found to be 8.60g. Find the mass of ferrous sulphate formed in this reaction. **[IJSO 2017-2018]**

(A) 19.40 g (B) 18.40 g (C) 17.40 g (D) 16.40 g

48. The ratio of atoms present in 4 g of magnesium and 4 g of sulphur is (Mg = 24; S = 32) **[IJSO 2017-2018]**

(A) 3: 4 (B) 2: 1 (C) 1: 1 (D) 3: 2

49. Arun needs 1.71 g of cane sugar ($C_{12}H_{22}O_{11}$) to sweeten his tea. What would be the number of carbon atoms consumed through sugar in the tea? **[IJSO 2017-2018]**

(A) 3.66×10^{22} (B) 7.2×10^{21} (C) 5×10^{21} (D) 6.6×10^{22}

50. A substance is dissolved in water, forming a 0.5 molar solution. If 4.0 L of solution contains 240g of the substance, what is the molecular mass of the substance? **[IJSO 2018-2019]**

(A) 60 g/mole (B) 120 g/mole (C) 240 g/mole (D) 480 g/mole

51. A car battery was kept for charging and after getting fully charged density of the battery acid (H_2SO_4) was measured and found to be 1.28 g cm^{-3}. If Initial molarity of battery acid was 4.2 M then mass percentage will be around _____. **[IJSO 2018-2019]**

(A) 28% (B) 30% (C) 32% (D) 34%

52. 4.095×10^{24} nitrogen atoms are filled in an enclosed gas cylinder of capacity two litre. The number of moles of nitrogen gas in the cylinder is _____. **[IJSO 2018-2019]**

(A) 14.7 (B) 6.8 (C) 3.4 (D) 2.9

53. A solution of pure aluminium sulphate containing 0.170g of aluminium ions is treated with excess of barium hydroxide solution. Total weight of the precipitate will be: **[IJSO 2018-2019]**

(A) 0.5g (B) 2.7g (C) 1.7 g (D) 0.54g

54. Sulphur di-oxide gas and ammonia gas were mixed in different proportions. The pair of gases containing same number of molecules at NTP is _____. **[IJSO 2018-2019]**

(A) 1120 cm^3 of SO_2 + 0.85 g of ammonia (B) 0.25g mole of SO_2 + 2240 cm^3 of ammonia

(C) 1680 cm^3 of SO_2 +1.7 g of ammonia (D) 0.25 g mole of SO_2 + 0.85 g of ammonia

55. W g of pure coal was combusted in pure dry oxygen. The carbon dioxide gas obtained was absorbed in 0.1 M KOH solution. The complete absorption of CO_2 required 5 cm^3 of 0.1 M KOH. The amount of coal combusted is **[IJSO 2018-2019]**

(A) 3 mg (B) 6 mg (C) 11 mg (D) 12 mg

56. A strip of iron with mass 15.5 g is placed in a solution containing 21.0 g copper sulphate. After some time the reaction stops. Iron strip was found to have mass 8.5 g. The mass of copper formed was found to be 8.60 g. Find the mass of ferrous sulphate formed in this reaction. **[IJSO 2018-2019]**

(A) 19.40 g (B) 18.40 g (C) 17.40 g (D) 16.40 g

57. Four gram of mixture of calcium carbonate and sand is treated with excess of HCl and 0.880 g of carbon-di-oxide is produced. What is the percentage of calcium carbonate in original mixture? **[IJSO 2019-2020]**

(A) 40 % (B) 50% (C) 55% (D) 45%

58. A magician performed following act: He dipped Rs. 50 note in a 50% solution of alcohol in water and held it on the burning flame, but the note did not burn. The reason behind this is – **[IJSO 2019-2020]**

(A) The alcohol kept on dousing the fire

(B) Air required for burning was not available

(C) The Rs. 50 note failed to reach ignition temperature

(D) The Rs. 50 note is fire proof

59. A zinc rod was dipped in 100 cm^3 of 1M copper chloride solution. After certain time the molarity of Cu^{2+} ions in the solution was found to be 0.8 M. If the eight of zinc rod is 20 g, then the molarity of chloride ions is_____. **[IJSO 2019-2020]**

(A) 2 M (B) 1.5 M (C) 1 M (D) 0.5 M

60. What is the percentage of $MgCO_3$ in a mixture of $MgCO_3$ and $CaCO_3$ if its 2g require 2g H_2SO_4 for complete neutralization? **[IJSO 2022]**

(A) 89% (B) 11% (C) 50% (D) 25%

61. An element X has two natural isotopes: ${}^{10}_{5}X$ (atomic mass 10.013 u) and ${}^{11}_{5}X$ (atomic mass 11.009 u). Relative abundance of these isotopes in nature has been recorded 19.8 % and 80.2 % respectively. On the basis of these data, average atomic mass of element X is close to **[IJSO 2023]**

(A) 10.210 u (B) 10.511 u (C) 10.799 u (D) 10.812 u

62. A mass 0.75 g of the mixture of Na_2CO_3 and K_2CO_3 is completely neutralized by 50 mL 0.25 N HCl. The percentage of Na_2CO_3 in the mixture is: **[IJSO 2023]**

(A) 50.6 (B) 49.4 (C) 50 (D) data insufficient

63. A boy gifted a diamond ring to his mother on her wedding anniversary. If this diamond ring contains 3 carat diamond then number of carbon atoms he gifted to his mother is: **[IJSO 2023]**
Given – (1 carat = 200 mg)
(A) 3.01×10^{23} (B) 2.1×10^{23} (C) 3.01×10^{22} (D) 2.1×10^{22}

64. Find the number of moles of hydrogen gas liberated when 39 g of potassium is treated with 7.8 g of water. **[INJSO 2023]**
(A) 0.22 mol (B) 0.43 mol (C) 0.50 mol (D) 1.0 mol

65. A closed container has a mixture of 48 g of sodium hydroxide, 52 g of water and 132 g ammonium sulphate. Find the number of moles of oxygen atoms present in that container. **[INJSO 2023]**
(A) 5 (B) 7 (C) 8 (D) 10

IJSO Stage-II

1. What amount of oxygen, O_2, (in moles) contains 1.8×10^{22} molecules? **[IJSO 2008-2009]**
(A) 0.0030 (B) 0.030 (C) 0.30 (D) 3.0

2. At the same Temperature and Pressure which sample contains the same number of moles of particles as 1 liter of O_2 (g)? **[IJSO 2008-2009]**
(A) 1L Ne (g) (B) 2L N_2 (g) (C) 0.5L SO_2 (g) (D) 1L H_2O (g)

3. Which compound has the empirical formula with the greatest mass? **[IJSO 2008-2009]**
(A) C_2H_6 (B) C_4H_{10} (C) C_5H_{10} (D) C_6H_6

4. Which of the following combination of elements of given atomic number can lead to a compound with a chemical formula of XY_3? **[IJSO 2008-2009]**
(A) 2 and 6 (B) 5 and 15 (C) 3 and 18 (D) 13 and 17

5. Which sample at STP has the same number of molecules as 5 liters of NO_2 (g) at STP? **[IJSO 2008-2009]**
(A) 5 grams of H_2 (g) (B) 5 liters of CH_4 (g)
(C) 5 moles of O_2 (g) (D) 5×10^{23} molecules of CO_2 (g)

6. Given 1 dm^3 of 0.15 M HCl and 1 dm^3 of 0.40 M HCl. What is the maximum volume of 0.25 M HCl which can be made by directly mixing the two solutions without adding water. **[IJSO 2009-2010]**
(A) 1 dm^3 (B) 2 dm^3 (C) 0.667 dm^3 (D) 1.667 dm^3

7. Calculate the mass of Lithium that contains same number of atoms as present in 8g of Magnesium. Atomic masses of lithium and magnesium are 7 and 24 respectively. **[IJSO 2009-2010]**
(A) 8g (B) 3g (C) 7g (D) 2.3 g

8. The volume occupied by 1.6 g of oxygen under NTP conditions is: **[IJSO 2009-2010]**
(A) 1.12 dm^3 (B) 5.60 dm^3 (C) 11.20 dm^3 (D) 22.4 dm^3

9. What will be the volume of Cl_2 at STP produced during electrolysis of $MgCl_2$ which produces 6.5 g Mg (At. wt. of Mg = 24.3g, Cl = 35.5g). **[IJSO 2010-2011]**
(A) 5.099 litre (B) 5.99 litre (C) 12.02 litre (D) 3.099 litre

10. A certain sample of concentrated hydrochloric acid contains 50% HCl by mass and has density 1.20 gcm^{-3}. What is the molarity of this sample? **[IJSO 2010-2011]**
(A) 16.4 M (B) 8.2 M (C) 32.8 M (D) 13.4 M

11. The density of water at room temperature is 1 g/ml (milliliter). Consider a spherical drop of water having volume 0.05ml. The drop evaporates at a uniform rate in one hour. The number of molecules leaving the liquid surface per second is approximately **[IJSO 2012-2013]**
(A) Zero (B) 5×10^{17} (C) 3×10^{14} (D) 2×10^{21}

12. 50 ml of 0.20 M solution of washing soda reacts with one of the acids in aqua regia. One of the products is Chile saltpeter. If the strength of the acid is 0.25 M, what volume of it will be required to react with the above solution? **[IJSO 2012-2013]**
(A) 40 ml. (B) 10 ml. (C) 20 ml. (D) 80 ml.

13. 1.84 g of Dolomite $CaMg(CO_3)^2$ ore was heated resulting in a residue of constant weight 0.96 g. During heating the metal of one of the product burnt with a dazzling white flame. The approximate percentage composition of the two products in the residue are respectively **[IJSO 2012-2013]**

(A) 54 and 46 (B) 46 and 54 (C) 42 and 58 (D) 58 and 42

14. Haber's process for the production of ammonia is an industrially important process used mainly in fertilizer industry. Due to some accident only 10% conversion took place. If all the volumes are measured in liters at STP, what volume of reactants are needed for this conversion? **[IJSO 2012-2013]**

(A) 1/2 of the total volume of hydrogen and nitrogen.
(B) $1/5^{th}$ of the total volume of hydrogen and nitrogen
(C) $1/10^{th}$ of the total volume of hydrogen and nitrogen
(D) $1/5^{th}$ of total volume of nitrogen

15. X is a member of alkene series with a molecular mass 28 amu. 200 cm^3 X is burnt in just sufficient air (containing 20% oxygen) to form carbon dioxide and steam. If all the measurements are made at constant pressure and 100º C, find the composition of the products formed and the unreacted air.

[IJSO 2012-2013]

(A) 400 cm^3, 400 cm^3, 600 cm^3 respectively (B) 200 cm^3, 200cm^3, 2400cm^3 respectively
(C) 200 cm^3, 400 cm^3, 2400 cm^3 respectively (D) 400 cm^3, 400 cm^3, 2400 cm^3 respectively

16. The amount of $CaCO_3$ which will precipitate if 50ml of 1.0M Na_2CO_3 and 50ml of 0.2M $CaCl_2$ are mixed is: **[IJSO 2013-2014]**

(A) 5.0 g (B) 2.0 g (C) 1.0 g (D) 0.5 g

17. How many molecules of water of hydration are present in 252mg of oxalic acid ($H_2C_2O_4.2H_2O$)?

[IJSO 2013-2014]

(A) 2.68×10^{18} (B) 2.52×10^{21} (C) 1.83×10^{24} (D) 2.4×10^{21}

18. Steel is an alloy of carbon and iron which is used in many utensils. Iron extracted will be of two types pig iron and wrought iron. Iron is extracted from its ore hematite by reduction with carbon monoxide. What (approximate) volume of carbon monoxide at NTP will be required to reduce 31.94 kg of hematite?

[IJSO 2014-2015]

(A) 12275L (B) 4480L (C) 9953L (D) 13440L

19. A gas is obtained by fractional distillation of liquid air. The same gas can also be obtained by Haber's process at high temperature and low pressure. Find out the number of atoms present in one liter of the gas

[IJSO 2014-2015]

(A) 2.67×10^{22} (B) 1.056×10^{23} (C) 5.28×10^{22} (D) 2.136×10^{23}

20. Sudha was studying the reaction between aluminium and iodine using 1.2 g aluminium and 2.4 g iodine. Calculate the weight of aluminium iodide formed. **[IJSO 2014-2015]**

(A) 2.57g (B) 2.49g (C) 1.35 g (D) 1.25g

21. The composition (v/v) of air is found to be 78% nitrogen, 21% oxygen, and 1% argon. The density of air at STP is **[IJSO 2015-2016]**

(A) 2.24 mg/cm^3 (B) 0.65 mg/cm^3 (C) 0.39mg/cm^3 (D) 1.3 mg/cm^3

22. Suhita took out 100cm^3 of X molar aqueous solution of hydrogen peroxide from the refrigerator. On warming to 19.5ºC, she observed 3dm^3 of oxygen gas was produced (at 1 atm pressure). Assuming complete decomposition of hydrogen peroxide, the value of X is approximately **[IJSO 2015-2016]**

(A) 1.25 (B) 2.5 (C) 0.8 (D) 0.5

23. Miska combined on liter oxygen gas and one liter hydrogen gas by applying an electric spark in a vessel at STP. She observed that water is formed. Find the weight of water and the weight of unreacted component respectively. **[IJSO 2015-2016]**

(A) 0.4017 g and 0.7143 g (B) 0.8036 g and 1.4286 g

(C) 0.4017 g and 1.4286 g (D) 0.8036 g and 0.7143 g

24. A black dot has a mass of about one femto gram. Assuming that the dot is made up of carbon only, calculate the approximate number of carbon atoms present in the dot. **[IJSO 2016-2017]**

(A) 1×10^8 (B) 5×10^{10} (C) 5×10^7 (D) 1×10^{11}

25. When ad mixture of 60mL of carbon monoxide and 40 mL of oxygen is sparked, mixture A is obtained. Mixture A is passed through aqueous potassium hydroxide to yield mixture B. The volumes of mixtures A and B respectively are **[IJSO 2016-2017]**

(A) 70 mL, 10 mL (B) 40 mL, 20 mL (C) 60 ml, 20 mL (D) 80 mL, 10 mL

26. A solution of pure ferric sulphate containing 0.140g of ferric ions is treated with excess of barium hydroxide solution. Total weight of the precipitate will be. **[IJSO 2016-2017]**

(A) 0.87 g (B) 1.14 g (C) 0.25 g (D) 0.56 g

27. Natural water contains about 0.02% D_2O (heavy water). When it is enriched to 20% (by volume) calculate the fraction of weight due to neutrons in 1 mole of water sample. **[IJSO 2017-2018]**

(A) 0.355 (B) 0.444 (C) 0.455 (D) 0.500

28. A piece of alloy of Al and Zn weighing 1.67 g was completely dissolved in acid and evolved 1.69 litre of hydrogen at NTP. What is the percentage of Al in the piece of alloy? **[IJSO 2017-2018]**

(A) 26.5% (B) 48.5% (C) 51.5% (D) 73.5%

29. An astronaut has to burn 40 g of glucose in his body per hour to get the required energy. Find the amount of oxygen that would need to be carried in space to meet his energy requirement for thirty days. **[IJSO 2017-2018]**

(A) 10.2 kg (B) 28.8 kg (C) 30.7 kg (D) 96.1 kg

30. A sample of clay was partially dried and then found to contain 60% silica and 8% water. The original sample of clay contained 15% water. Find the percentage of silica in the original sample. **[IJSO 2017-2018]**

(A) 52.3% (B) 47.8% (C) 55.5% (D) 51.7%

31. How many molecules of water of hydration are present in 252 mg of oxalic acid ($C_2H_2O_4.2H_2O$)? **[IJSO 2017-2018]**

(A) 1.2×10^{21} (B) 1.7×10^{21} (C) 2.4×10^{21} (D) 3.4×10^{21}

32. One spoon of a sample of common salt weighs approximately 0.5 g. It contains 40% sodium and 380 micrograms of iodine. Assuming that the sample contains only sodium, iodide and chloride ion, the number of chloride ions present in one spoon of this sample is closest to **[IJSO 2018-2019]**

(A) 5×10^{20} (B) 5×10^{21} (C) 5×10^{22} (D) 5×10^{23}

33. An LPG gas cylinder regularly used in the household contains a, mixture of butane and propane. If 5 litres of this mixture on complete combustion produces 17 litres of CO_2 at atmospheric pressure and 25°C, then the ratio of butane to propane in the mixture is (Assume that both the gases in the cylinder are in vapour phase.) **[IJSO 2018-2019]**

(A) 3 : 2 (B) 2 : 3 (C) 4 : 1 (D) 1 : 4

34. Sachin was suffering from problem of acidity, so he visited a physician who advised him to take 0.025 dm^3 of milk of magnesia for a fast relief. He exactly followed what the doctor told him to do. Out of curiosity he saw the label on milk of magnesia bottle and he found that there were different ingredients written on it and the concentration of milk of magnesia mentioned was 29 ppm.

Assuming, the volume of milk of magnesia required for neutralization of acid is equal to intake of milk of magnesia, help Sachin to find out the following: **[IJSO 2011-2012]**

i. How many moles of acid was produced in Sachin's stomach?

ii. Write down the neutralization reaction of this process.

iii. Calculate the concentration of acid produced in mol/dm^3.

35. The mass percent of MnO_2 in a sample of a mineral is determined by reacting it with a measured excess of As_2O_3 in acid solution, and then titrating the remaining As_2O_3 with standard $KMnO_4$. A 0.225 g sample of the mineral is ground and boiled with 75.0 mL of 0.0125 M As_2O_3 solution containing 10mL of concentrated sulfuric acid. After the reaction is complete, the solution is cooled, diluted with water, and titrated with 2.25×10^{-3}M $KMnO_4$, requiring 16.00ml to reach the endpoint. **[IJSO 2013-2014]**

Note: 5 mol of As_2O_3 react with 4 mol of MnO_4^-

i. Write a balanced equation for the reaction of As_2O_3 with MnO_2 in acid solution. The products are Mn^{2+} and AsO_4^{3-}.

ii. Calculate the number of moles of

  i. As_2O_3 added initially.

  ii. MnO_4^- used to titrate the excess As_2O_3.

  iii. MnO_2 in the sample.

iii. Determine the mass percent of MnO_2 in the sample.

iv. Describe how the endpoint is selected in the $KMnO_4$ titration

36. The purpose of an air bag is to slow the passenger's forward movement into the steering wheel (or dash board) during a collision and also to provide a cushion between the passenger and the steering wheel. The goal of an air bag is to help the passenger come to a stop with minimum damage. One of the ways an air bag helps reduce injury is by spreading the force of impact with the dashboard or steering wheel over a larger area, as illustrated in Figure 1. When the force is spread over a larger area of the body, the injuries are less severe. **[IJSO 2013-2014]**

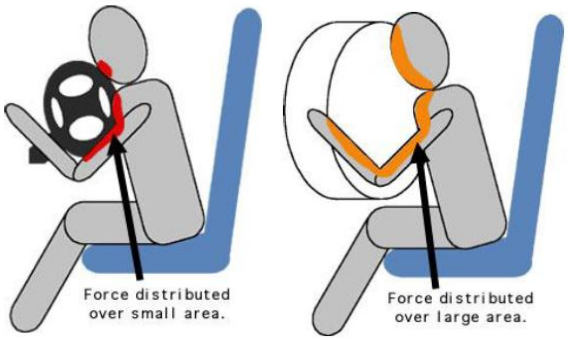


Fig.: Force distribution during collision comparing air bag to no air bag. **[IJSO 2015-2016]**

A certain model of car is equipped with 65.0 liter air bag that inflates to 89.4 m/s in 40 milliseconds. The weight of air bag is 2.0 Kg and the thickness of fully inflated air bag is 30.0 cm.

One method used to inflate air bags in cars is to use nitrogen produced chemically from the decomposition of sodium azide:

$2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$ ———— (1)

The sodium formed reacts with potassium nitrate to give more nitrogen:

$10\ Na(s) + 2KNO_3(s) \longrightarrow K_2O(s) + 5Na_2O(s) + N_2(g)$ ———— (2)

i. Calculate the ratio (by mass) in which the sodium azide and potassium nitrate should be mixed in order that no metallic sodium remains after the reaction.

Na_2O and K_2O, are highly reactive, so it would be unsafe to allow them to be the end product of the airbag detonation. These metal oxides react with silicon dioxide (SiO_2) in a final reaction to produce silicate glass, which is harmless and stable.

ii. Write the reaction of SiO_2 with sodium oxide and potassium oxide.

iii. Calculate the total mass of the solid mixture of sodium azide and potassium nitrate needed to inflate a 72 dm^3 air bag is filled with nitrogen gas at of 1 atm and at room temperature (27 ºC).

Consider the molar volume of nitrogen gas as 24.0 dm^3 at 300 K and universal gas constant, R = 0.0821 liter. atm. $mole^{-1}K^{-1}$

(Important: Show all your calculation step clearly)

The sodium azide is prepared commercially by the reaction between dinitrogen monoxide and sodium amide.

$N_2O(g) + 2NaNH_2(s) \longrightarrow NaN_3(s) + NaOH(s) + NH_3(g)\ \Delta H_r = 55.8\ KJ / mol$

iv. Calculate ΔH_r for reaction (1) above, the decomposition of sodium azide.

Given:

Compound	$N_2O(g)$	$NaNH_2(s)$	$NaOH(s)$	$NH_3(g)$
ΔH_f (KJ/mol)	+82.0	–123.7	–425.2	–46.1

37. Jiya had a piece of plumber's solder weighing 3.0g. She dissolved it in dilute nitric acid and the resultant solution she treated with dil.H_2SO_4. She got precipitate of lead sulphate which after washing and drying weighed 2.93g. From the filtrate she obtained stannic acid. She heated stannic acid to obtain stannic oxide. The yield of stannic oxide was 1.27 g. Help Jiya to find out percentage of lead and stannous in the piece of plumber's solder.

38. Potassium chlorate ($KClO_3$) decomposes on heating into potassium chloride (KCl) and oxygen (O_2) gas. Potassium bicarbonate ($KHCO_3$) decomposes on heating to give potassium carbonate (K_2CO_3) and releases water vapour and carbon dioxide. Potassium carbonate (K_2CO_3) on further heating decomposes to potassium oxide (K_2O) and carbon dioxide. However, potassium chloride (KCl) does not decompose on further heating.

Dina has a mixture of potassium chlorate, potassium bicarbonate and potassium carbonate. When she heated 1000g of the mixture, she observed that 18g water vapour, 290g carbon dioxide and 40g oxygen gas. Assuming complete decomposition, what is the composition of the initial mixture in weight%?

39. **(A)** A hydrocarbon contained 10.5g of carbon per gm of hydrogen. One litre of the hydrocarbon vapours at 127 °C and 1 atm pressure weighed 2.8g. Help Sakshi to find the molecular formula of the hydrocarbon.

(B) 40 mL of mixture of hydrogen and oxygen gases was placed in a gas burette at 48°C and 1 atm pressure. A spark was applied so that the formation of water is complete. The volume of the remaining hydrogen gas was 10 ml at 48°C and 1 atm pressure. Find the initial mole percentage of hydrogen in the mixture. **[IJSO 2016-2017]**

40. During respiration, glucose combines with oxygen in the cells of our body to form carbon dioxide, water and production of energy. Atmospheric air consists of 20 % oxygen by volume, only 5% of that oxygen is consumed by the body in each breath of an average person at STP. According to the study an average person at rest inhales 8 litres of air per minute. In 3.5 hrs, how much glucose is used up by an average person at rest? Find the amount of carbon dioxide exhaled during the process. **[IJSO 2016-2017]**

41. Veeni thermally decomposed 90g of potassium chlorate of 60% purity. The total gas produced was allowed to react with hydrogen that was prepared by passing steam over hot magnesium metal. Calculate the amount magnesium required to produce just sufficient hydrogen for completion of the reaction.
[IJSO 2017-2018]

42. Deepa has 100 ml of 10.7% ammonium chloride solution. Vikram added some amount of slaked lime in it. The gas produced in the reaction is allowed to react with 23.85 g of copper (II) oxide to give solid copper. Calculate: **[IJSO 2017-2018]**

(i) Write balanced chemical equations for all the reactions involved

(ii) The amount of slaked lime required to produce sufficient amount of the gas for complete reaction with copper oxide.

(iii) The mass of copper after the reaction.

43. The molecular formula of a gaseous compound is to be determined. This compound is found to be composed of 85.7% by mass carbon and 14.3% by mass hydrogen. Its density is 2.28 gL^{-1} at 300 K and 1 atm. pressure. Form the given data **[IJSO 2018-2019]**

(A) Calculate the number of moles of carbon atoms present in 100 g of compound.
Show your calculations in the box.
Answer: _____.

(B) Calculate the number of moles of hydrogen atoms present in 100 g of compound.
Show your calculations in the box.
Answer: _____.

(C) The empirical formula of the compound is: _____

(D) Moles / Litre of the compound at NTP = _____.
Show your calculations in the box .

(E) Empirical formula units = _____.

(F) Molecular formula = _____.

Answer Key

EXERCISE #1

DALTON'S ATOMIC THEORY

1. (D) **2.** (C) **3.** (B)

LAWS OF CHEMICAL COMBINATION

1. (B) **2.** (A) **3.** (C) **4.** (D)
5. (A) **6.** (B) **7.** (B) **8.** (C)
9. (C) **10.** (C) **11.** (A) **12.** (C)
13. (B) **14.** (C) **15.** (B)

PERCENTAGE COMPOSITION

1. (B) **2.** (A) **3.** (A) **4.** (A)
5. (A) **6.** (D) **7.** (C) **8.** (C)
9. (C) **10.** (B)

MOLE CONCEPT

1. (C) **2.** (B) **3.** (C) **4.** (D)
5. (C) **6.** (D) **7.** (B) **8.** (A)
9. (B) **10.** (A) **11.** (C) **12.** (D)
13. (B) **14.** (A) **15.** (C) **16.** (A)
17. (D) **18.** (A) **19.** (B) **20.** (B)
21. (C) **22.** (B) **23.** (A) **24.** (D)
25. (B)

LIMITING REAGENT

1. (B) **2.** (C) **3.** (C) **4.** (A)
5. (B) **6.** (A) **7.** (B)

CONCENTRATION TERMS

1. (C) **2.** (A) **3.** (B) **4.** (B)
5. (B) **6.** (D) **7.** (B) **8.** (B)
9. (A) **10.** (A) **11.** (C) **12.** (A)
13. (B) **14.** (B) **15.** (A) **16.** (B)
17. (A) **18.** (B) **19.** (B) **20.** (D)

pH OF SOLUTION

1. (B) **2.** (C) **3.** (D) **4.** (B)
5. (B) **6.** (B) **7.** (A) **8.** (A)
9. (C) **10.** (A) **11.** (C) **12.** (D)
13. (B) **14.** (D) **15.** (A)

EXERCISE #2

1. (A) **2.** (B) **3.** (D) **4.** (C)
5. (C) **6.** (A) **7.** (C) **8.** (D)
9. (A) **10.** (C) **11.** (A) **12.** (D)
13. (D) **14.** (C) **15.** (D) **16.** (A)
17. (B) **18.** (C) **19.** (C) **20.** (D)
21. (A) **22.** (D) **23.** (C) **24.** (A)
25. (C) **26.** (B) **27.** (D) **28.** (C)
29. (C) **30.** (D) **31.** (D) **32.** (C)
33. (A) **34.** (C) **35.** (C) **36.** (D)
37. (A) **38.** (B) **39.** (A) **40.** (D)
41. (B) **42.** (C) **43.** (A) **44.** (D)
45. (B) **46.** (C) **47.** (A) **48.** (D)
49. (D) **50.** (D) **51.** (A) **52.** (A)
53. (D) **54.** (D) **55.** (A) **56.** (A)
57. (D) **58.** (B) **59.** (A) **60.** (B)
61. (C) **62.** (B) **63.** (B) **64.** (C)
65. (C) **66.** (D) **67.** (D) **68.** (D)
69. (A) **70.** (B) **71.** (B) **72.** (B)
73. (B) **74.** (A) **75.** (C) **76.** (B)
77. (A) **78.** (B) **79.** (A) **80.** (B)
81. (C) **82.** (C) **83.** (B) **84.** (C)
85. (B)

NSEJS PYQ

1. (C) **2.** (B) **3.** (D) **4.** (A)
5. (A) **6.** (B) **7.** (C) **8.** (C)
9. (A) **10.** (B) **11.** (D) **12.** (B)
13. (C) **14.** (A) **15.** (C) **16.** (B)
17. (A) **18.** (A) **19.** (A) **20.** (A)
21. (C) **22.** (A) **23.** (B) **24.** (C)
25. (B) **26.** (A) **27.** (B) **28.** (A)
29. (C) **30.** (D) **31.** (A) **32.** (A)
33. (C) **34.** (A) **35.** (A, B, D) **36.** (C)
37. (A) **38.** (A) **39.** (D) **40.** (B)
41. (C) **42.** (B) **43.** (B) **44.** (B)
45. (D) **46.** (C) **47.** (C) **48.** (D)
49. (B) **50.** (B) **51.** (A)

Advanced & Olympiad Challenger Questions

IJSO STAG I

1. (B)	**2.** (B)	**3.** (C)	**4.** (A)
5. (C)	**6.** (D)	**7.** (A)	**8.** (C)
9. (B)	**10.** (B)	**11.** (B)	**12.** (A)
13. (A)	**14.** (D)	**15.** (B)	**16.** (A)
17. (C)	**18.** (C)	**19.** (C)	**20.** (C)
21. (B)	**22.** (C)	**23.** (A)	**24.** (B)
25. (D)	**26.** (D)	**27.** (B)	**28.** (C)
29. (B)	**30.** (B)	**31.** (A)	**32.** (C)
33. (B)	**34.** (D)	**35.** (A)	**36.** (D)
37. (B)	**38.** (A)	**39.** (C)	**40.** (B)
41. (A)	**42.** (B)	**43.** (C)	**44.** (A)
45. (B)	**46.** (A)	**47.** (A)	**48.** (A)
49. (C)	**50.** (A)	**51.** (B)	**52.** (C)
53. (C)	**54.** (B)	**55.** (A)	**56.** (A)
57. (B)	**58.** (C)	**59.** (A)	**60.** (B)
61. (D)	**62.** (B)	**63.** (C)	**64.** (C)
65. (A)			

IJSO STAG II

1. (B)	**2.** (D)	**3.** (B)	**4.** (D)
5. (B)	**6.** (D)	**7.** (D)	**8.** (A)
9. (B)	**10.** (A)	**11.** (B)	**12.** (A)
13. (D)	**14.** (C)	**15.** (D)	**16.** (C)
17. (D)	**18.** (D)	**19.** (C)	**20.** (A)
21. (D)	**22.** (B)	**23.** (D)	**24.** (C)
25. (A)	**26.** (B)	**27.** (C)	**28.** (D)
29. (C)	**30.** (C)	**31.** (C)	**32.** (B)
33. (B)			

34. i. 0.0005 mol dm^3

ii. $Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + H_2O$

iii. 1.25×10^4

35. i. $2MnO_2 + As_2O_3 + H_2O \rightarrow 2Mn^{2+} + 2AsO_4^{3-} + 2H^+$

ii. i. 9.38×10^{-4} mol As_2O_3 ii. 3.6×10^{-5} mol MnO_4^-, 4.5×10^{-5} mol As_2O_3 left

iii. 1.8×10^{-5} mol MnO_2

iii. 62 % MnO_2 in sample.

iv. The endpoint corresponds to a slight purple (pink) color due to excess MnO_4^- (aq).

36. i. 3.22: 1 (mass by ratio)

ii. $K_2O + Na_2O + SiO_2 \longrightarrow Na_2K_2SiO_4$ (alkaline silicate glass)

iii. 159.4g

iv. –723.4 KJ/mol

37. 66.7 % of Pb, 33.3 % of Sn

38. Composition of the original mixture is: 10.2% chlorate, 20% bicarbonate and 69.8% carbonate.

39. **A.** C_7H_8. **B.** 75%

40. Glucose is used up = 22.5 g, Amount of carbon dioxide exhaled = 33 g.

41. 31.8 g of Mg

42. (i) $Ca(OH)_2$ (s) + $2NH_4Cl$(aq) $\rightarrow CaCl_2$ (aq) + $2NH_3$ (g) + $2H_2O$ (l)
(ii) Amount of slaked lime required is 7.4 g
(iii) Amount of Copper obtained is 19.05 g

43. [A] 7.12 moles of carbon
[B] 14.24 moles of hydrogen
[C] Empirical formula = $(CH_2)_n$
[D] 0.0408 mol/L
[E] Empirical formula unit = CH_2
[F] molecular formula = C_4H_8

